

Bài thực hành số 2 — Web3.py & Ape Framework

> Geth trên Windows thường dùng HTTP `http://127.0.0.1:8545` thay vì IPC `/tmp/geth.ipc` (Linux/Mac). IPC Windows (nếu bật): `\\.\pipe\geth.ipc`.

A — Web3.py + Ganache

i. Thiết lập môi trường

Cài Web3.py

```
In [2]: %pip install web3
```

Collecting web3
Using cached web3-8.0.0-py3-none-any.whl.metadata (3.7 kB)
Collecting aiohttp>=3.7.4.post0 (from web3)
Using cached aiohttp-3.14.4-cp312-cp312-win_amd64.whl.metadata (8.5 kB)
Collecting eth-abi>=6 (from web3)
Using cached eth_abi-6.0.0-py3-none-any.whl.metadata (2.4 kB)
Collecting eth-account>=0.14 (from web3)
Using cached eth_account-0.14.0-py3-none-any.whl.metadata (2.4 kB)
Collecting eth-hash>=0.5.1 (from eth-hash[pycryptodome]>=0.5.1->web3)
Using cached eth_hash-0.8.0-py3-none-any.whl.metadata (4.1 kB)
Collecting eth-typing>=5.0.0 (from web3)
Using cached eth_typing-6.0.0-py3-none-any.whl.metadata (3.1 kB)
Collecting eth-utils>=5.3.0 (from web3)
Using cached eth_utils-6.0.0-py3-none-any.whl.metadata (5.6 kB)
Collecting hexbytes>=2 (from web3)
Using cached hexbytes-2.0.0-py3-none-any.whl.metadata (2.1 kB)
Collecting pydantic>=2.4 (from web3)
Using cached pydantic-2.13.5-py3-none-any.whl.metadata (110 kB)
Collecting pyunormalize>=15 (from web3)
Using cached pyunormalize-18.0.0-py3-none-any.whl.metadata (5.5 kB)
Collecting pywin32>=223 (from web3)
Using cached pywin32-312-cp312-cp312-win_amd64.whl.metadata (11 kB)
Collecting requests>=2.23.0 (from web3)
Using cached requests-2.34.2-py3-none-any.whl.metadata (4.8 kB)
Collecting websockets>=14 (from web3)
Downloading websockets-17.2-cp312-cp312-win_amd64.whl.metadata (6.5 kB)
Collecting aiohappyeyeballs>=2.5.0 (from aiohttp>=3.7.4.post0->web3)
Using cached aiohappyeyeballs-2.7.1-py3-none-any.whl.metadata (5.9 kB)
Collecting aiosignal>=1.4.0 (from aiohttp>=3.7.4.post0->web3)
Using cached aiosignal-1.4.0-py3-none-any.whl.metadata (3.7 kB)
Collecting attrs>=17.3.0 (from aiohttp>=3.7.4.post0->web3)
Using cached attrs-26.1.0-py3-none-any.whl.metadata (8.8 kB)
Collecting frozenlist>=1.1.1 (from aiohttp>=3.7.4.post0->web3)
Using cached frozenlist-1.8.0-cp312-cp312-win_amd64.whl.metadata (21 kB)
Collecting multidict<7.0,>=4.5 (from aiohttp>=3.7.4.post0->web3)
Using cached multidict-6.9.1-cp312-cp312-win_amd64.whl.metadata (5.4 kB)
Collecting propcache>=0.2.0 (from aiohttp>=3.7.4.post0->web3)
Using cached propcache-0.5.4-cp312-cp312-win_amd64.whl.metadata (25 kB)
Requirement already satisfied: typing_extensions>=4.4 in d:\Study\IS355\venv-ape\Lib\site-packages (from aiohttp>=3.7.4.post0->web3) (4.16.0)
Collecting yarl<2.0,>=1.25.1 (from aiohttp>=3.7.4.post0->web3)
Using cached yarl-1.25.1-cp312-cp312-win_amd64.whl.metadata (115 kB)
Collecting idna>=2.0 (from yarl<2.0,>=1.25.1->aiohttp>=3.7.4.post0->web3)
Using cached idna-3.20-py3-none-any.whl.metadata (7.2 kB)
Collecting parsimonious<0.11.0,>=0.10.0 (from eth-abi>=6->web3)
Using cached parsimonious-0.10.0-py3-none-any.whl.metadata (25 kB)
Collecting regex>=2022.3.15 (from parsimonious<0.11.0,>=0.10.0->eth-abi>=6->web3)
Using cached regex-2026.9.29-cp312-cp312-win_amd64.whl.metadata (41 kB)
Collecting bitarray>=2.4 (from eth-account>=0.14->web3)
Using cached bitarray-3.11.0-cp312-cp312-win_amd64.whl.metadata (39 kB)
Collecting eth-keyfile>=0.10 (from eth-account>=0.14->web3)
Using cached eth_keyfile-0.10.0-py3-none-any.whl.metadata (9.8 kB)
Collecting eth-keys>=0.5 (from eth-account>=0.14->web3)
Using cached eth_keys-0.8.0-py3-none-any.whl.metadata (12 kB)
Collecting eth-rlp>=3 (from eth-account>=0.14->web3)
Using cached eth_rlp-3.0.0-py3-none-any.whl.metadata (2.2 kB)

Collecting rlp>=2 (from eth-account>=0.14->web3)
Using cached rlp-5.0.0-py3-none-any.whl.metadata (2.1 kB)
Collecting ckzg>=2 (from eth-account>=0.14->web3)
Using cached ckzg-2.1.8-cp312-cp312-win_amd64.whl.metadata (930 bytes)
Requirement already satisfied: pycryptodome<4,>=3.6.6 in d:\Study\IS355\venv-ape\Li
b\site-packages (from eth-hash[pycryptodome]>=0.5.1->web3) (3.24.0)
Collecting py_ecc>=5.2.0 (from eth-keyfile>=0.10->eth-account>=0.14->web3)
Using cached py_ecc-8.0.0-py3-none-any.whl.metadata (3.1 kB)
Collecting cytoolz>=0.10.1 (from eth-utils>=5.3.0->web3)
Using cached cytoolz-1.1.0-cp312-cp312-win_amd64.whl.metadata (5.3 kB)
Collecting annotated-types>=0.6.0 (from pydantic>=2.4->web3)
Using cached annotated_types-0.8.0-py3-none-any.whl.metadata (15 kB)
Collecting pydantic-core==2.46.5 (from pydantic>=2.4->web3)
Using cached pydantic_core-2.46.5-cp312-cp312-win_amd64.whl.metadata (6.7 kB)
Collecting typing-inspection>=0.4.2 (from pydantic>=2.4->web3)
Using cached typing_inspection-0.4.4-py3-none-any.whl.metadata (2.6 kB)
Collecting toolz>=0.8.0 (from cytoolz>=0.10.1->eth-utils>=5.3.0->web3)
Using cached toolz-1.1.0-py3-none-any.whl.metadata (5.1 kB)
Collecting charset_normalizer<4,>=2 (from requests>=2.23.0->web3)
Using cached charset_normalizer-3.5.2-cp312-cp312-win_amd64.whl.metadata (47 kB)
Collecting urllib3<3,>=1.26 (from requests>=2.23.0->web3)
Using cached urllib3-2.8.0-py3-none-any.whl.metadata (7.4 kB)
Collecting certifi>=2023.5.7 (from requests>=2.23.0->web3)
Using cached certifi-2026.7.22-py3-none-any.whl.metadata (2.5 kB)
Using cached web3-8.0.0-py3-none-any.whl (1.4 MB)
Using cached aiohttp-3.14.4-cp312-cp312-win_amd64.whl (506 kB)
Using cached multidict-6.9.1-cp312-cp312-win_amd64.whl (59 kB)
Using cached yarl-1.25.1-cp312-cp312-win_amd64.whl (102 kB)
Using cached aiohappyeyeballs-2.7.1-py3-none-any.whl (15 kB)
Using cached aiosignal-1.4.0-py3-none-any.whl (7.5 kB)
Using cached attrs-26.1.0-py3-none-any.whl (67 kB)
Using cached eth_abi-6.0.0-py3-none-any.whl (53 kB)
Using cached parsimonious-0.10.0-py3-none-any.whl (48 kB)
Using cached eth_account-0.14.0-py3-none-any.whl (586 kB)
Using cached bitarray-3.11.0-cp312-cp312-win_amd64.whl (171 kB)
Using cached ckzg-2.1.8-cp312-cp312-win_amd64.whl (103 kB)
Using cached eth_hash-0.8.0-py3-none-any.whl (8.0 kB)
Using cached eth_keyfile-0.10.0-py3-none-any.whl (9.5 kB)
Using cached eth_keys-0.8.0-py3-none-any.whl (20 kB)
Using cached eth_rlp-3.0.0-py3-none-any.whl (15 kB)
Using cached eth_typing-6.0.0-py3-none-any.whl (19 kB)
Using cached eth_utils-6.0.0-py3-none-any.whl (102 kB)
Using cached pydantic-2.13.5-py3-none-any.whl (472 kB)
Using cached pydantic_core-2.46.5-cp312-cp312-win_amd64.whl (2.0 MB)
Using cached annotated_types-0.8.0-py3-none-any.whl (13 kB)
Using cached cytoolz-1.1.0-cp312-cp312-win_amd64.whl (947 kB)
Using cached frozenlist-1.8.0-cp312-cp312-win_amd64.whl (44 kB)
Using cached hexbytes-2.0.0-py3-none-any.whl (18 kB)
Using cached idna-3.20-py3-none-any.whl (69 kB)
Using cached propcache-0.5.4-cp312-cp312-win_amd64.whl (46 kB)
Using cached py_ecc-8.0.0-py3-none-any.whl (47 kB)
Using cached pyunormalize-18.0.0-py3-none-any.whl (78 kB)
Using cached pywin32-312-cp312-cp312-win_amd64.whl (6.9 MB)
Using cached regex-2026.9.29-cp312-cp312-win_amd64.whl (280 kB)
Using cached requests-2.34.2-py3-none-any.whl (73 kB)
Using cached charset_normalizer-3.5.2-cp312-cp312-win_amd64.whl (207 kB)


```
----- 6/38 [pywin32]
----- 6/38 [pywin32]
----- 6/38 [pywin32]
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----- 6/38 [pywin32]
----- 6/38 [pywin32]
----- 6/38 [pywin32]
----- 6/38 [pywin32]
----- 6/38 [pywin32]
----- 7/38 [pyunormalize]
----- 7/38 [pyunormalize]
----- 9/38 [propcache]
----- 11/38 [idna]
----- 11/38 [idna]
----- 14/38 [eth-typing]
----- 15/38 [eth-hash]
----- 16/38 [charset_normalizer]
----- 16/38 [charset_normalizer]
----- 18/38 [bitarray]
----- 19/38 [attrs]
----- 19/38 [attrs]
----- 21/38 [aiohappyeyeballs]
----- 23/38 [requests]
----- 23/38 [requests]
----- 24/38 [pydantic]
----- 24/38 [pydantic]
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----- 24/38 [pydantic]
----- 24/38 [pydantic]
----- 24/38 [pydantic]
----- 24/38 [pydantic]
----- 24/38 [pydantic]
----- 25/38 [parsimonious]
----- 25/38 [parsimonious]
----- 26/38 [cytoolz]
----- 26/38 [cytoolz]
----- 28/38 [eth-utils]
----- 28/38 [eth-utils]
----- 29/38 [aiohttp]
----- 29/38 [aiohttp]
----- 29/38 [aiohttp]
----- 29/38 [aiohttp]
----- 29/38 [aiohttp]
```

[illegible]

Successfully installed aiohappyeyeballs-2.7.1 aiohttp-3.14.4 aiosignal-1.4.0 annotated-types-0.8.0 attrs-26.1.0 bitarray-3.11.0 certifi-2026.7.22 charset_normalizer-3.5.2 ckzg-2.1.8 cytoolz-1.1.0 eth-abi-6.0.0 eth-account-0.14.0 eth-hash-0.8.0 eth-keyfile-0.10.0 eth-keys-0.8.0 eth-rlp-3.0.0 eth-typing-6.0.0 eth-utils-6.0.0 frozenlist-1.8.0 hexbytes-2.0.0 idna-3.20 multidict-6.9.1 parsimonious-0.10.0 propcache-0.5.4 py-ecc-8.0.0 pydantic-2.13.5 pydantic-core-2.46.5 pyunormalize-18.0.0 pywin32-312 regex-2026.9.29 requests-2.34.2 rlp-5.0.0 toolz-1.1.0 typing-inspection-0.4.4 urllib3-2.8.0 web3-8.0.0 websockets-17.2 yarl-1.25.1

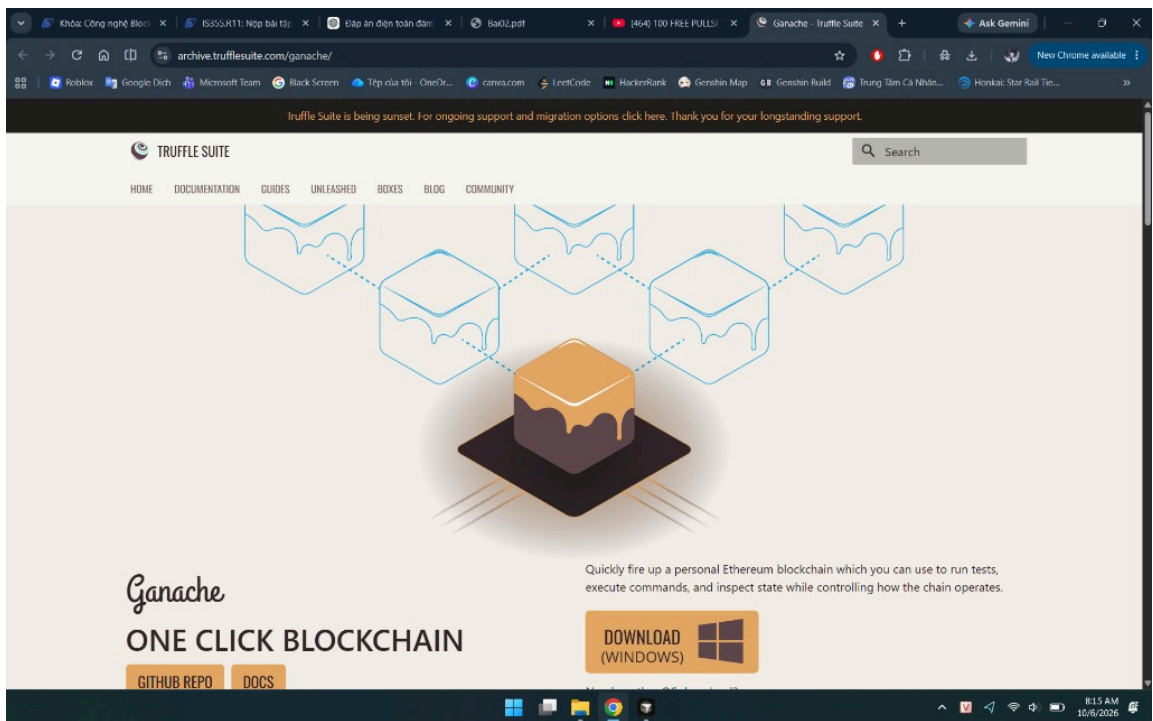
Note: you may need to restart the kernel to use updated packages.

Cài / mở Ganache

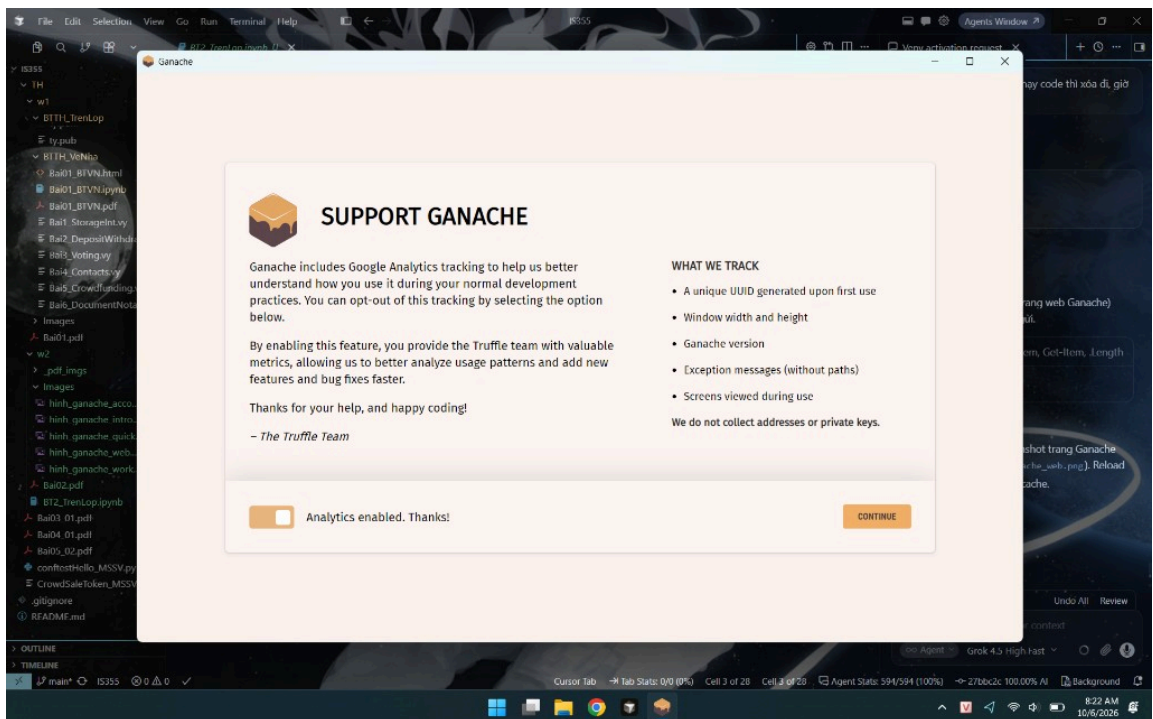
- Tải: <https://archive.trufflesuite.com/ganache/>
- Tạo **New Workspace** (ví dụ tên **IS355-BC**), bấm **Start**.
- RPC server: thường **http://127.0.0.1:7545** .

Ảnh minh họa

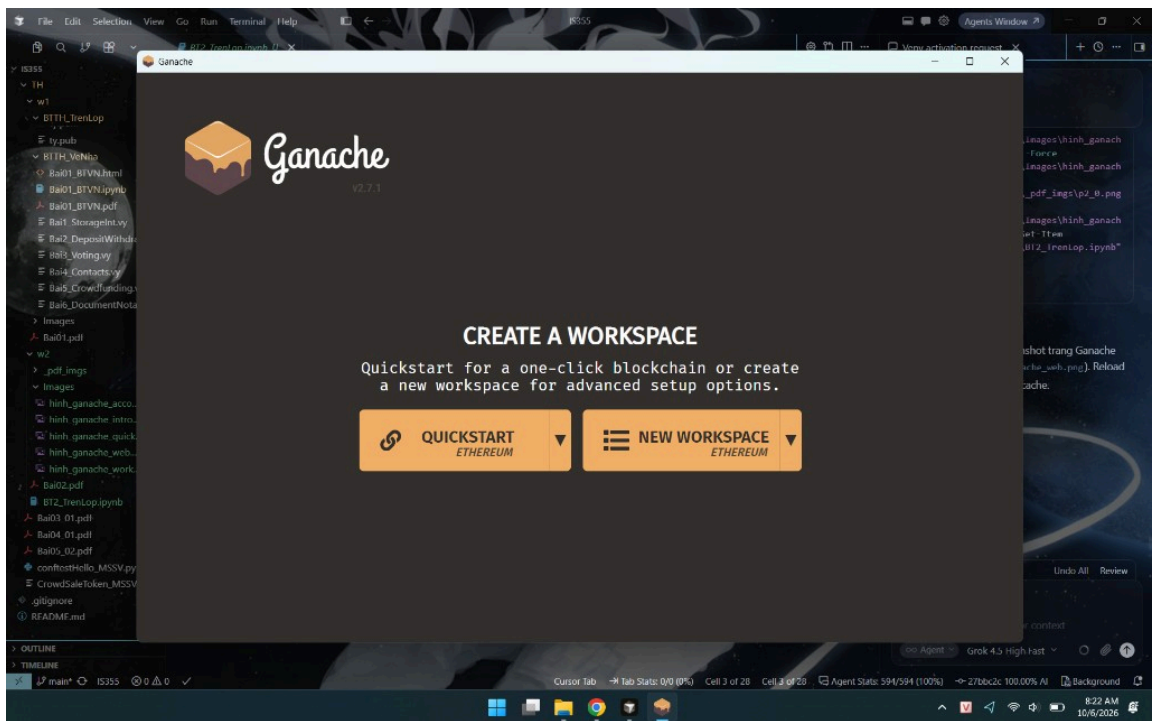
Hình 1 — trang web Ganache:



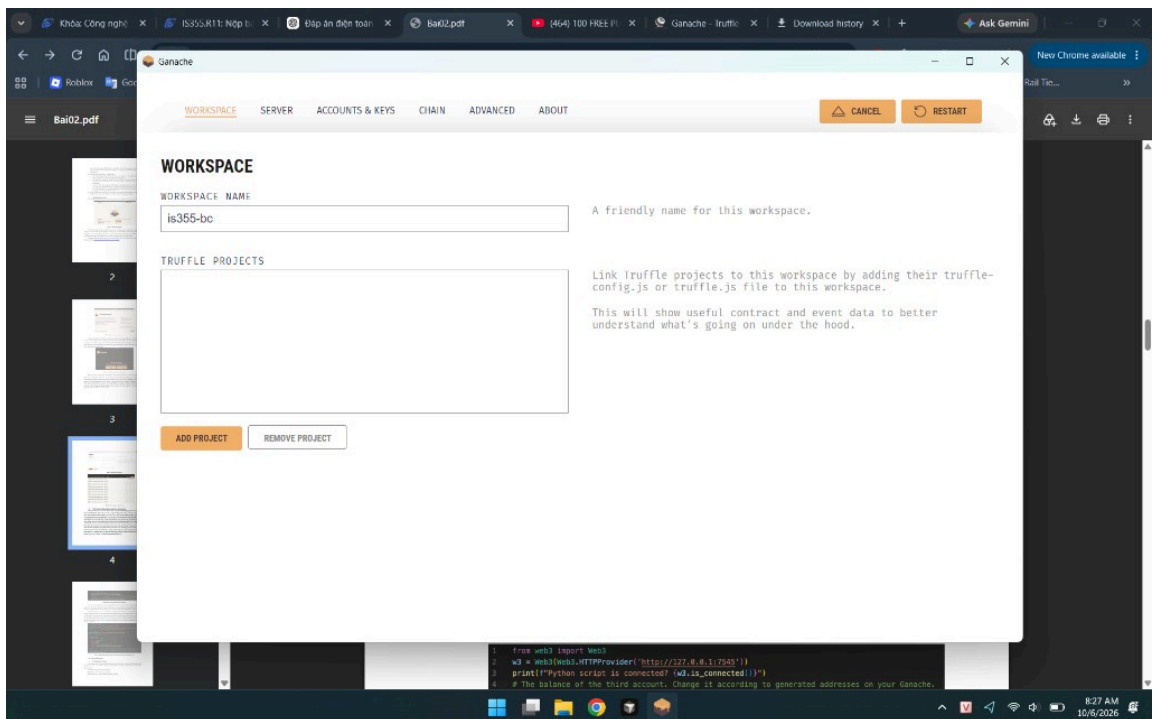
Hình 2 — màn hình giới thiệu:



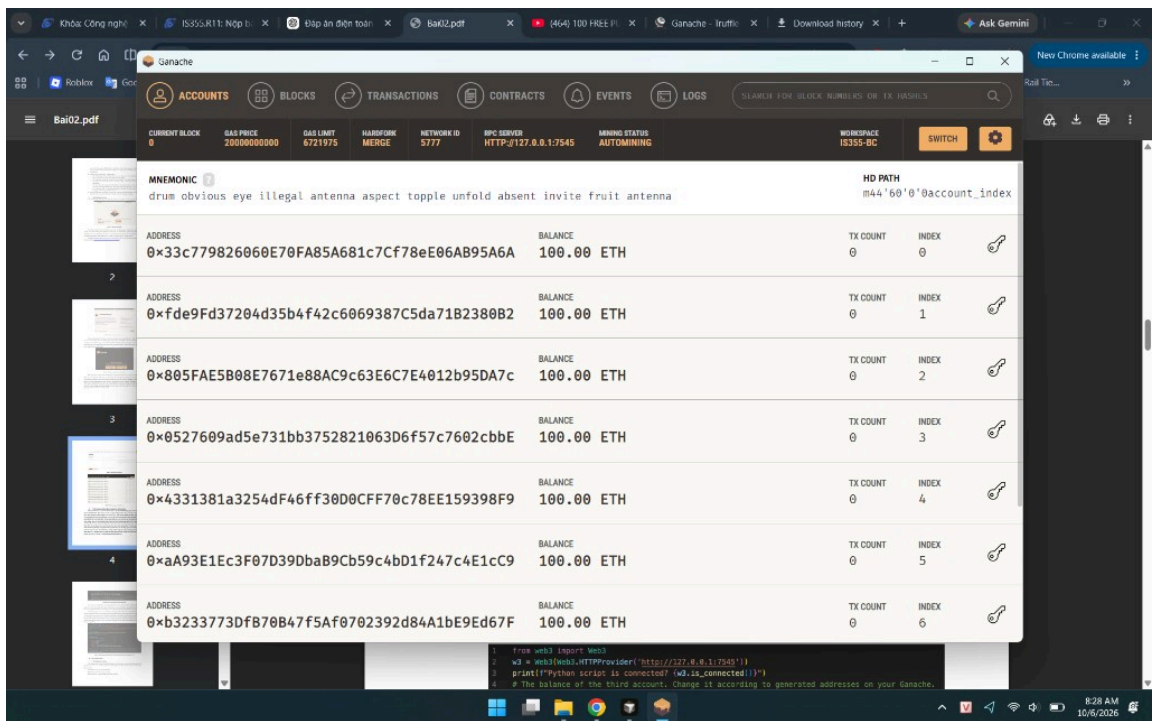
Hình 3 — Quickstart:



Hình 4 — Workspace:



Hình 5 — Accounts (mỗi account ~100 ETH, block 0):



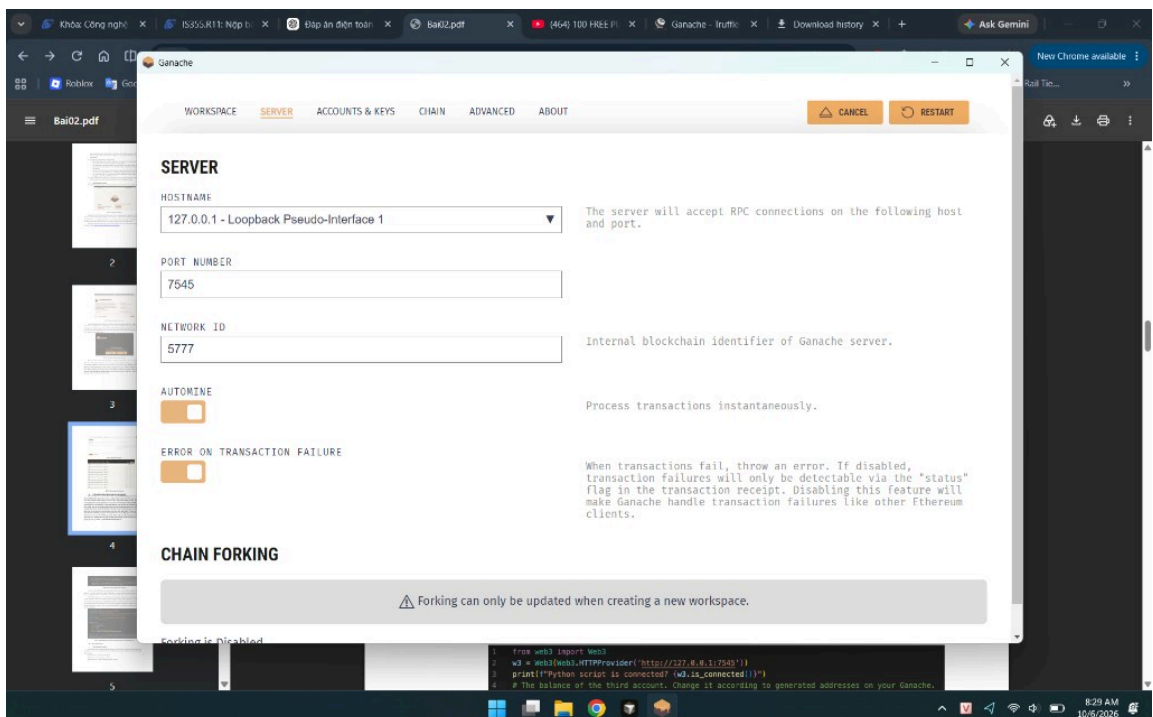
Yêu cầu báo cáo (Ganache tabs)

Chụp và mô tả ngắn các tab: **Server, Accounts & keys, Chain, Advanced, About.**

Các tab cấu hình Ganache (workspace IS355-BC)

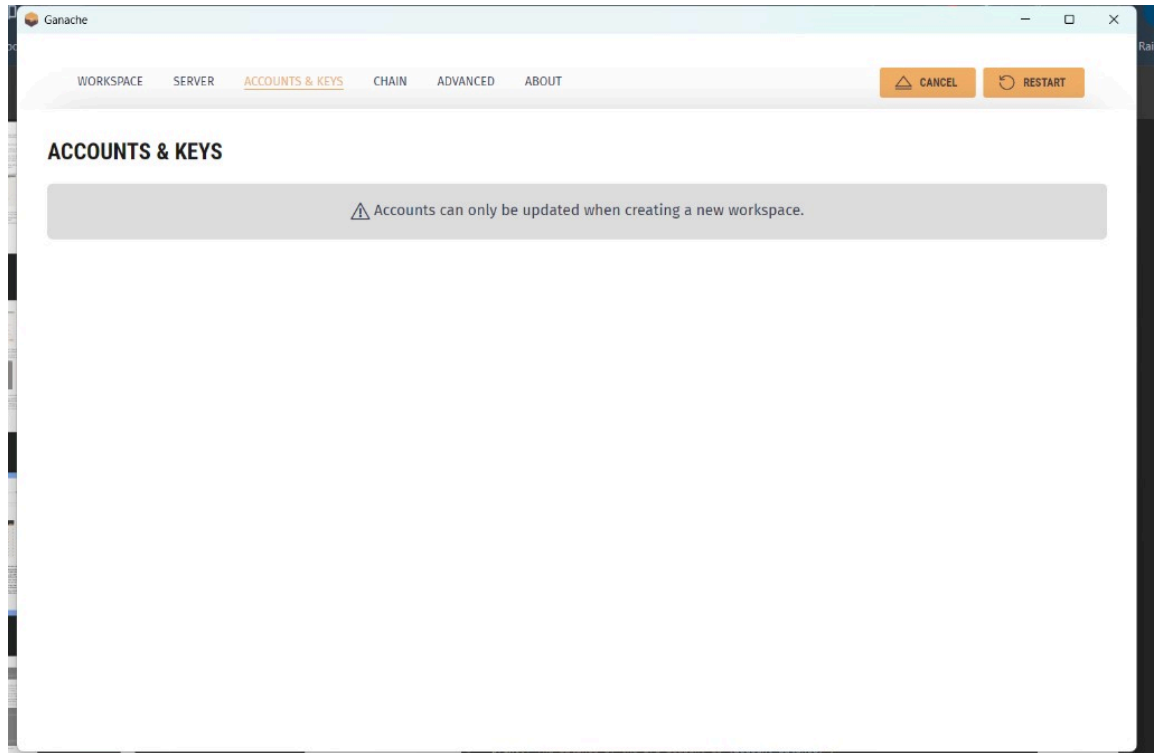
Yêu cầu đề: chụp và giới thiệu các tab **Server, Accounts & keys, Chain, Advanced, About.**

Tab Server



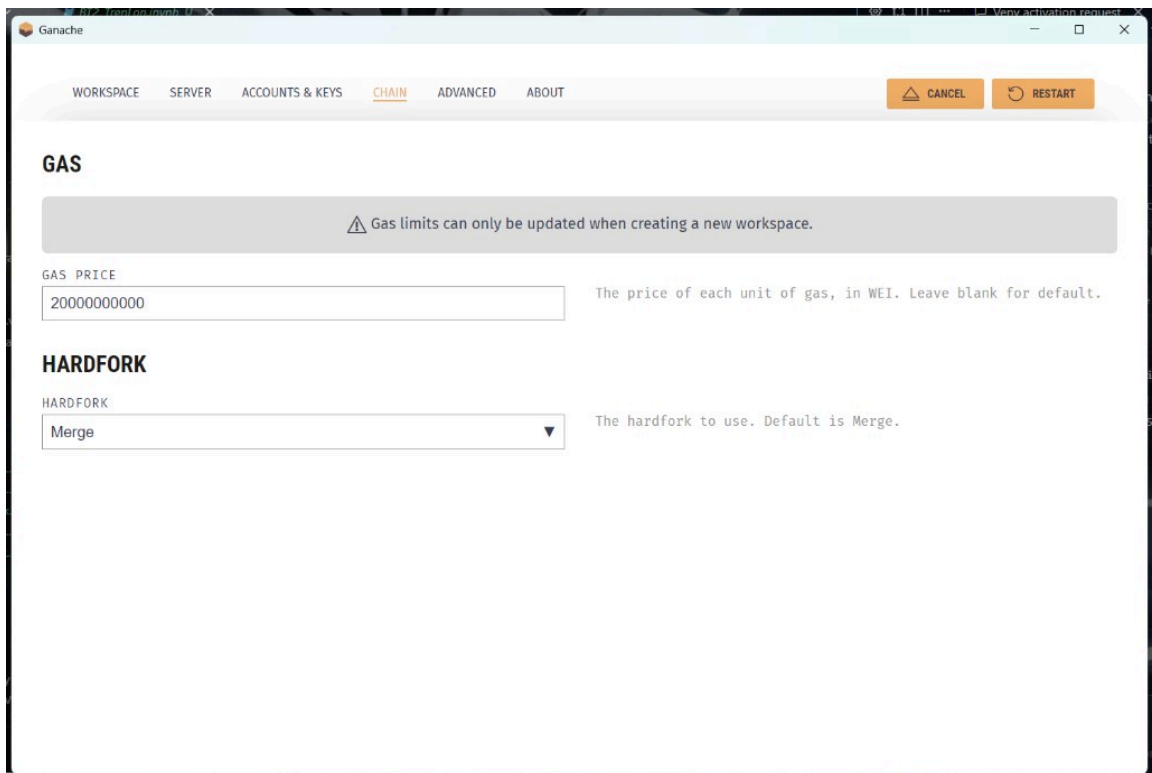
- **Hostname:** 127.0.0.1
- **Port:** 7545 → RPC http://127.0.0.1:7545
- **Network ID:** 5777
- **Automine:** bật
- **Error on transaction failure:** bật
- Chain forking: disabled (chỉ đổi khi tạo workspace mới)

Tab Accounts & Keys



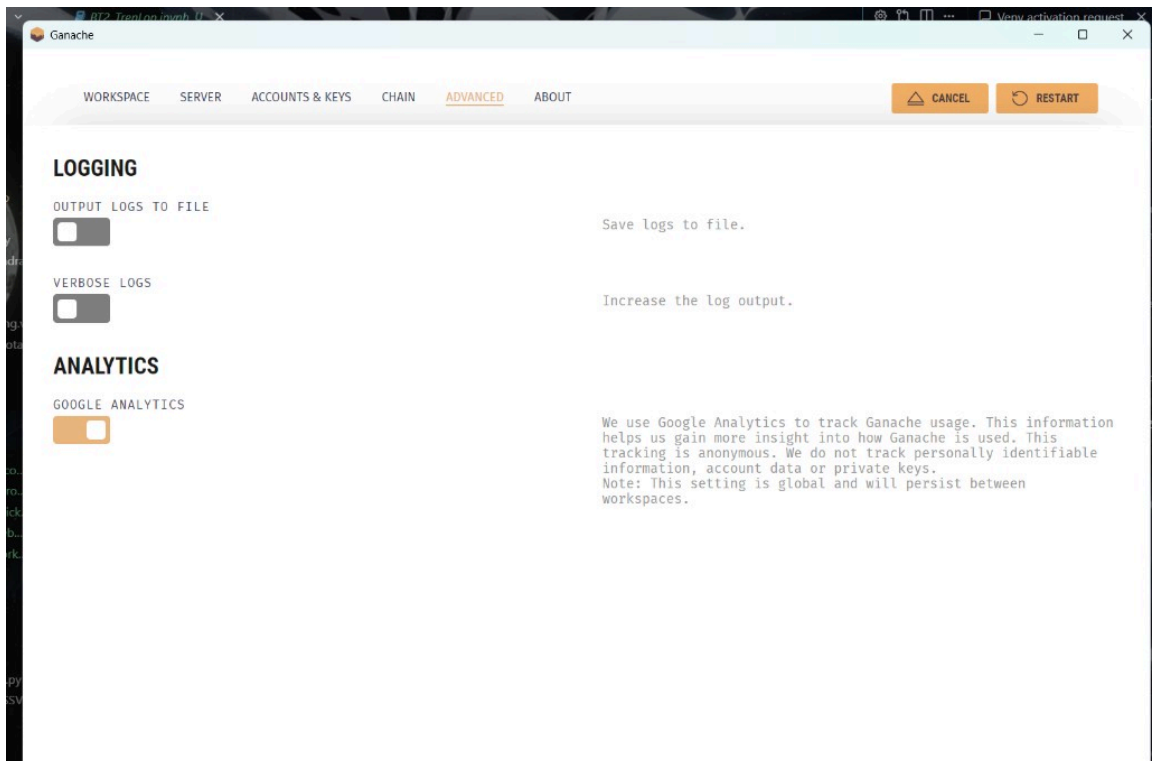
- Với workspace đã tạo: **không sửa được** accounts/keys (thông báo *Accounts can only be updated when creating a new workspace*).
- Muốn đổi số account / mnemonic / ETH ban đầu → tạo workspace mới.
- Danh sách account xem ở màn hình Accounts chính (Hình 5).

Tab Chain



- **Gas Price:** 20000000000 wei (= 20 gwei)
- **Hardfork:** Merge
- Gas limit chỉ đổi được lúc tạo workspace mới

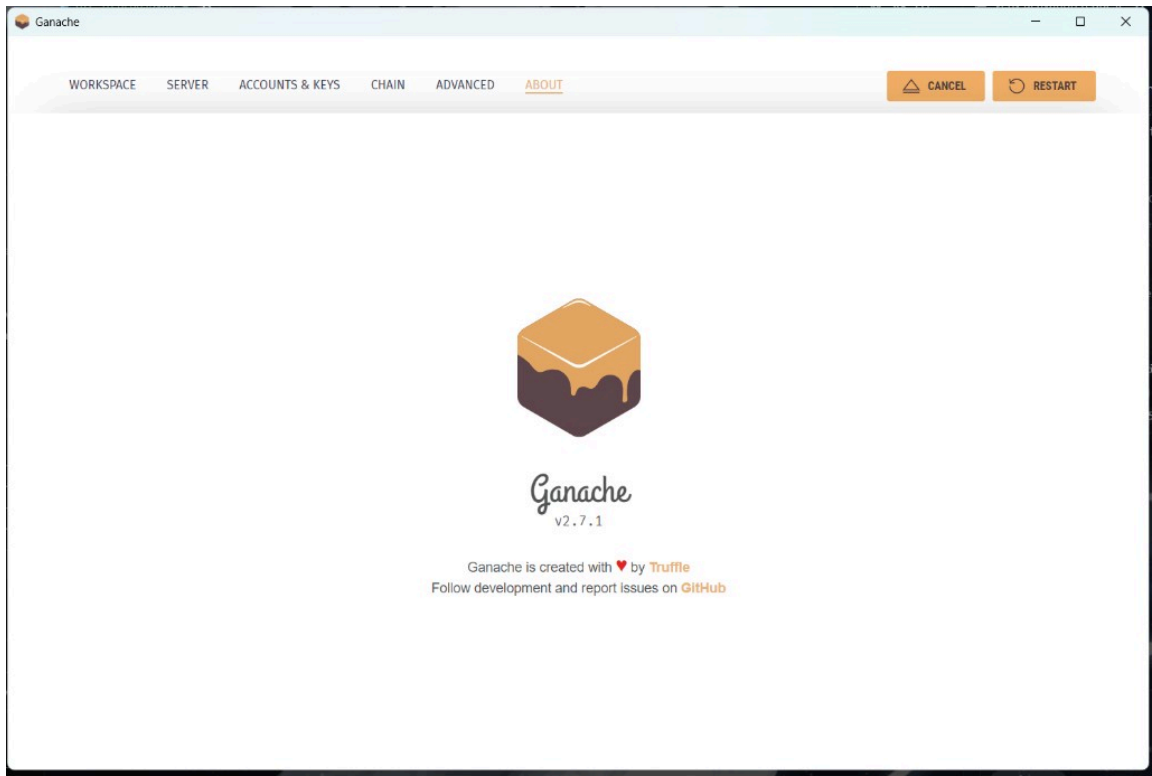
Tab Advanced



- **Output logs to file:** tắt

- **Verbose logs:** tắt
- **Google Analytics:** bật (global, không lưu address/private key)

Tab About



- **Version:** Ganache **v2.7.1** (Truffle)

Tab	Nội dung chính (máy này)
Server	127.0.0.1:7545 , Network ID 5777 , Automine ON
Accounts & keys	Chỉ xem được khi tạo workspace mới; account hiện tại xem Hình 5
Chain	Gas price 20 gwei , Hardfork Merge
Advanced	Logging OFF; Analytics ON
About	v2.7.1

ii. Tạo account + chuyển ETH

Bản cài Geth không có `clef.exe` → dùng `geth account new` (tạo keystore tương đương).

Cách làm

```
cd D:\Study\IS355\TH\w2
mkdir my_keystore -Force
D:\Tools\Geth\geth.exe account new --keystore my_keystore
```

```
PS D:\Study\IS355\TH\w2> cd D:\Study\IS355\TH\w2
PS D:\Study\IS355\TH\w2> mkdir my_keystore -Force

Directory: D:\Study\IS355\TH\w2

Mode                LastWriteTime         Length Name
----                -
d-----         10/6/2026  11:00 AM             my_keystore

PS D:\Study\IS355\TH\w2> D:\Tools\Geth\geth.exe account new --keystore my_keystore
INFO [10-06|11:03:43.800] Maximum peer count          ETH=50 total=50
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:  0x69f1AebA4Bf643A4C9533a6Fd5Eb64c8fC78fD5f
Path of the secret key file: D:\Study\IS355\TH\w2\my_keystore\UTC--2026-10-06T04-03-59.286226600Z--69f1aeba4bf643a4c9533a6fd5eb64c8fc78fd5f

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!
```

Máy này:

- Address: 0x69f1Aeba4Bf643A4C9533a6Fd5Eb64c8fC78fD5f
- Keystore: my_keystore/UTC--2026-10-06T04-03-59.286226600Z--69f1aeba4bf643a4c9533a6fd5eb64c8fc78fd5f
- Password: 123456 (dùng decrypt keystore/Python — **không** dán vào sendTransaction)

Chuyển 10 ETH trong console Geth

Không dùng "0xDIA_CHI_MOI" (placeholder). Dán địa chỉ thật:

```
eth.sendTransaction({from: eth.accounts[0], to:
"0x69f1Aeba4Bf643A4C9533a6Fd5Eb64c8fC78fD5f", value: web3.toWei(10,
"ether")})
web3.fromWei(eth.getBalance("0x69f1Aeba4Bf643A4C9533a6Fd5Eb64c8fC78fD5f"),
"ether")
```

Kết quả chuyển 10 ETH (balance = 10):

```
> eth.sendTransaction({from: eth.accounts[0], to: "0x69f1Aeba4Bf643A4C9533a6Fd5Eb64c8fC78fD5f", value: web3.toWei(10, "e
ther")})
"0xab73553b560df69451908ddeb410809cc6152233bed23afef0ffe33f0d4282d0"
> web3.fromWei(eth.getBalance("0x69f1Aeba4Bf643A4C9533a6Fd5Eb64c8fC78fD5f"), "ether")
10
```

Nếu dán nhầm placeholder → lỗi invalid address :

```

> eth.sendTransaction({from: eth.accounts[0], to: "0xDIA_CHI_MOI", value: web3.toWei(10, "ether")})
Error: invalid address
    at inputAddressFormatter (web3.js:3943:11(39))
    at inputTransactionFormatter (web3.js:3748:43(23))
    at web3.js:5049:37(8)
    at map (native)
    at web3.js:5048:35(12)
    at web3.js:5074:34(15)
    at send (web3.js:5099:39(11))
    at <eval>:1:20(16)

> web3.fromWei(eth.getBalance("0xDIA_CHI_MOI"), "ether")eth.sendTransaction({from: eth.accounts[0], to: "0xDIA_CHI_MOI",
value: web3.toWei(10, "ether")})
SyntaxError: SyntaxError: (anonymous): Line 1:55 Unexpected identifier (and 1 more errors)

```

```

In [3]: # hello_web3.py – kết nối Ganache & đọc balance
# Chạy cell này SAU KHI đã mở Ganache (RPC :7545)
from web3 import Web3

w3 = Web3(Web3.HTTPProvider("http://127.0.0.1:7545"))
print(f"Python script is connected? {w3.is_connected()}")

# Account INDEX 3 trên Ganache IS355-BC
ACCOUNT_3 = Web3.to_checksum_address("0x0527609ad5e731bb3752821063D6f57c7602cbbE")

account_balance = w3.eth.get_balance(ACCOUNT_3)
print(f"The balance of the 3rd account is {account_balance}")
print(f"Balance in ETH = {w3.from_wei(account_balance, 'ether')}")
print("Ghi nhớ: 1 ETH = 1_000_000_000_000_000_000 wei")

```

Python script is connected? True
The balance of the 3rd account is 1000000000000000000
Balance in ETH = 100
Ghi nhớ: 1 ETH = 1_000_000_000_000_000_000 wei

Private key account #3 (INDEX 3)

The screenshot shows the Ganache desktop application interface. A modal window titled "ACCOUNT INFORMATION" is open, displaying details for account #3. The background shows a list of accounts with their addresses and balances.

ACCOUNT ADDRESS	BALANCE
0x0527609ad5e731bb3752821063D6f57c7602cbbE	100.00 ETH
0x0527609ad5e731bb3752821063D6f57c7602cbbE	100.00 ETH
0x0527609ad5e731bb3752821063D6f57c7602cbbE	100.00 ETH
0x0527609ad5e731bb3752821063D6f57c7602cbbE	100.00 ETH
0x0527609ad5e731bb3752821063D6f57c7602cbbE	100.00 ETH
0x0527609ad5e731bb3752821063D6f57c7602cbbE	100.00 ETH
0x0527609ad5e731bb3752821063D6f57c7602cbbE	100.00 ETH
0x0527609ad5e731bb3752821063D6f57c7602cbbE	100.00 ETH
0x0527609ad5e731bb3752821063D6f57c7602cbbE	100.00 ETH

ACCOUNT INFORMATION

ACCOUNT ADDRESS
0x0527609ad5e731bb3752821063D6f57c7602cbbE

PRIVATE KEY
0x6552742db064c1cad2ebda4f5ba15a777999581d3a179d7e6c4a0b0950f61601

Do not use this private key on a public blockchain; use it for development purposes only!

DONE

- Address: 0x0527609ad5e731bb3752821063D6f57c7602cbbE
- Private key: (xem ảnh — **chỉ dùng trên Ganache local**)

iii. Chuyển ETH: account #3 → account #6

Các bước chương trình

1. Kết nối RPC Ganache.
2. Lấy `private_key` + `address` người gửi (#3), `address` người nhận (#6).
3. `amount = to_wei(...)`; lấy `nonce`.
4. Tạo dict `transaction ( from , to , value , gas , gasPrice , nonce )`.
5. `sign_transaction` → `send_raw_transaction` → (tùy chọn) chờ receipt.
6. In balance trước/sau để đối chiếu trên Ganache.

Bước 1 — Kết nối Ganache + khai báo account

```
In [6]: # transfer - account INDEX 3 -> INDEX 6
from web3 import Web3

w3 = Web3(Web3.HTTPProvider("http://127.0.0.1:7545"))
assert w3.is_connected(), "Mo Ganache truoc (http://127.0.0.1:7545)"

sender_private_key = "0x6552742db604c1cad2ebda4f5ba15a777999581d3a179d7e6c4a0b0950f"
sender_address = Web3.to_checksum_address("0x0527609ad5e731bb3752821063D6f57c7602cb")
recipient_address = Web3.to_checksum_address("0xb3233773DfB70B47f5Af0702392d84A1bE9")

# De bai: 5 wei. Hinh 7: doi thanh Web3.to_wei(5, "ether")
amount = Web3.to_wei(5, "wei")
print("connected:", w3.is_connected())
print("sender   :", sender_address)
print("recipient:", recipient_address)
print("amount   :", amount, "wei")
```

```
connected: True
sender   : 0x0527609ad5e731bb3752821063D6f57c7602cbbE
recipient: 0xb3233773DfB70B47f5Af0702392d84A1bE9Ed67F
amount   : 5 wei
```

Bước 2 — Balance trước khi chuyển

```
In [7]: print("=== BEFORE ===")
print("sender   :", w3.from_wei(w3.eth.get_balance(sender_address), "ether"), "ETH")
print("recipient:", w3.from_wei(w3.eth.get_balance(recipient_address), "ether"), "E")

=== BEFORE ===
sender   : 100 ETH
recipient: 100 ETH
```

Bước 3 — Tạo transaction (`gas` , `gasPrice` , `nonce`)

```
In [8]: transaction = {
    "from": sender_address,
    "to": recipient_address,
    "value": amount,
    "gas": 21000, # transfer thuong
    "gasPrice": Web3.to_wei("50", "gwei"),
    "nonce": w3.eth.get_transaction_count(sender_address),
}

print("=== TRANSACTION ===")
for k, v in transaction.items():
    print(f" {k}: {v}")
```

```
=== TRANSACTION ===
from: 0x0527609ad5e731bb3752821063D6f57c7602cbbE
to: 0xb3233773DfB70B47f5Af0702392d84A1bE9Ed67F
value: 5
gas: 21000
gasPrice: 50000000000
nonce: 0
```

Bước 4 — Ký tx + gửi lên chain

```
In [9]: signed = w3.eth.account.sign_transaction(transaction, sender_private_key)
raw = getattr(signed, "raw_transaction", None) or signed.rawTransaction
tx_hash = w3.eth.send_raw_transaction(raw)
receipt = w3.eth.wait_for_transaction_receipt(tx_hash)
print("tx hash:", tx_hash.hex())
print("status :", receipt["status"], "(1=success)")
print("gasUsed:", receipt["gasUsed"])
```

```
tx hash: 3e34b0a93aad3714166c95ecc51f786cf3700422c40c3a9d15e252ee84a53b29
status : 1 (1=success)
gasUsed: 21000
```

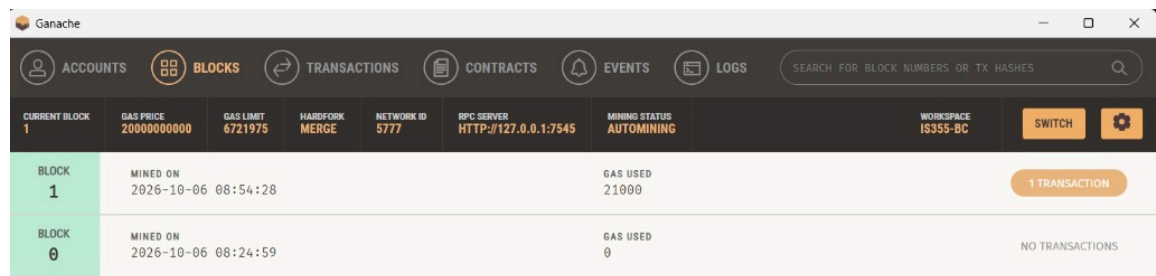
Bước 5 — Balance sau khi chuyển + phí gas

```
In [10]: print("=== AFTER ===")
print("tx hash :", tx_hash.hex())
print("status :", receipt["status"], "(1=success)")
print("gasUsed :", receipt["gasUsed"])
print("phi uoc tinh (wei):", receipt["gasUsed"] * transaction["gasPrice"])
print("sender   :", w3.from_wei(w3.eth.get_balance(sender_address), "ether"), "ETH")
print("recipient:", w3.from_wei(w3.eth.get_balance(recipient_address), "ether"), "E
```

```
=== AFTER ===
tx hash : 3e34b0a93aad3714166c95ecc51f786cf3700422c40c3a9d15e252ee84a53b29
status : 1 (1=success)
gasUsed : 21000
phi uoc tinh (wei): 1050000000000000
sender   : 99.998949999999999995 ETH
recipient: 100.00000000000000005 ETH
```

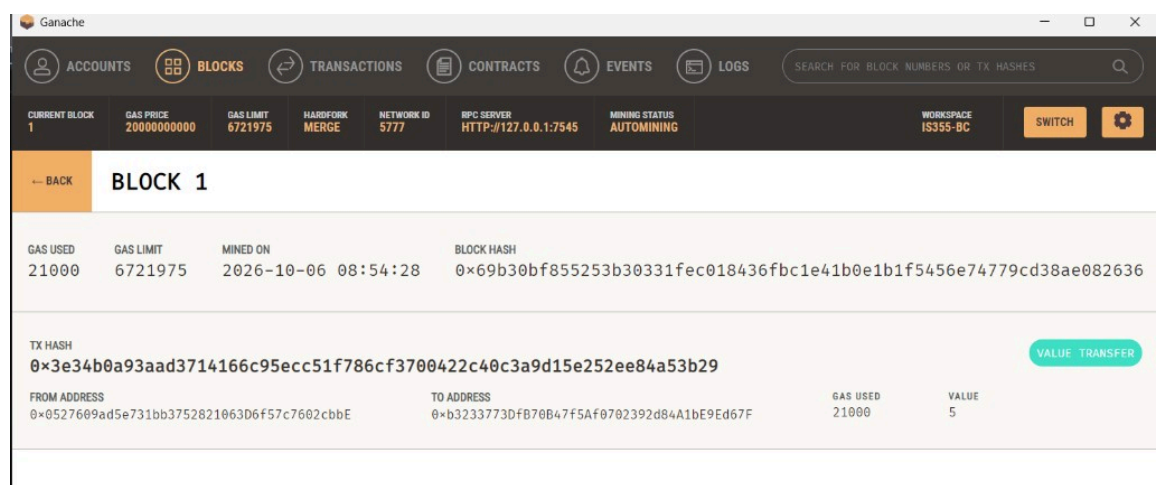
Kết quả trên Ganache (sau khi transfer)

Tab **BLOCKS** — Current Block = 1, Block 1 dùng **21000 gas**, có 1 transaction:



CURRENT BLOCK	GAS PRICE	GAS LIMIT	HARDFORK	NETWORK ID	RPC SERVER	MINING STATUS	WORKSPACE	
1	2000000000	6721975	MERGE	5777	HTTP://127.0.0.1:7545	AUTOMINING	IS355-BC	SWITCH
BLOCK 1	MINED ON 2026-10-06 08:54:28					GAS USED 21000		1 TRANSACTION
BLOCK 0	MINED ON 2026-10-06 08:24:59					GAS USED 0		NO TRANSACTIONS

Chi tiết **BLOCK 1** — VALUE TRANSFER 5 wei, INDEX 3 → INDEX 6:



BLOCK 1					
GAS USED 21000	GAS LIMIT 6721975	MINED ON 2026-10-06 08:54:28	BLOCK HASH 0x69b30bf855253b30331fec018436fbc1e41b0e1b1f5456e74779cd38ae082636		
TX HASH 0x3e34b0a93aad3714166c95ecc51f786cf3700422c40c3a9d15e252ee84a53b29					
FROM ADDRESS 0x0527609ad5e731bb3752821063D6f57c7602cbbE		TO ADDRESS 0xb3233773DfB70B47f5Af0702392d84A1bE9Ed67F		GAS USED 21000	VALUE 5

- **TX Hash:**
0x3e34b0a93aad3714166c95ecc51f786cf3700422c40c3a9d15e252ee84a53b29
- **From:** 0x0527609ad5e731bb3752821063D6f57c7602cbbE (INDEX 3)
- **To:** 0xb3233773DfB70B47f5Af0702392d84A1bE9Ed67F (INDEX 6)
- **Gas Used:** 21000 | **Value:** 5 wei

Giải thích nhanh gas & gasPrice

- **gas = 21000** : hạn mức tối đa cho simple transfer (khớp Gas Used trên Ganache).
- **gasPrice** : giá mỗi gas; trên Ganache automine thường không quan trọng thứ tự.
- **Phí thực tế** $\approx$ **gasUsed * gasPrice** = 21000 * 50 gwei .
- Sender mất thêm phí gas ngoài số ETH/wei chuyển.

B — Ape Framework

i. Cài đặt & `ape init`

Bước 1 — Cài `eth-ape`

```
In [1]: # Cài Ape Framework  
%pip install eth-ape
```

Collecting eth-ape
Using cached eth_ape-0.8.52-py3-none-any.whl.metadata (9.7 kB)
Collecting click<8.2,>=8.1.6 (from eth-ape)
Using cached click-8.1.8-py3-none-any.whl.metadata (2.3 kB)
Collecting ijson<4,>=3.1.4 (from eth-ape)
Downloading ijson-3.5.1-cp312-cp312-win_amd64.whl.metadata (24 kB)
Collecting ipython<9,>=8.18.1 (from eth-ape)
Using cached ipython-8.39.0-py3-none-any.whl.metadata (5.1 kB)
Collecting lazyasd>=0.1.4 (from eth-ape)
Using cached lazyasd-0.1.4.tar.gz (8.4 kB)
Installing build dependencies: started
Installing build dependencies: finished with status 'done'
Getting requirements to build wheel: started
Getting requirements to build wheel: finished with status 'done'
Preparing metadata (pyproject.toml): started
Preparing metadata (pyproject.toml): finished with status 'done'
Collecting asttokens<3,>=2.4.1 (from eth-ape)
Using cached asttokens-2.4.1-py2.py3-none-any.whl.metadata (5.2 kB)
Collecting cchecksum<1,>=0.0.3 (from eth-ape)
Downloading cchecksum-0.4.5-cp312-cp312-win_amd64.whl.metadata (2.3 kB)
Collecting numpy<2 (from eth-ape)
Downloading numpy-1.26.4-cp312-cp312-win_amd64.whl.metadata (61 kB)
Collecting packaging<24,>=23.0 (from eth-ape)
Downloading packaging-23.2-py3-none-any.whl.metadata (3.2 kB)
Collecting pandas<3,>=2.2.2 (from eth-ape)
Downloading pandas-2.3.3-cp312-cp312-win_amd64.whl.metadata (19 kB)
Collecting pluggy<2,>=1.3 (from eth-ape)
Downloading pluggy-1.6.0-py3-none-any.whl.metadata (4.8 kB)
Collecting pydantic<3,>=2.10 (from eth-ape)
Using cached pydantic-2.13.5-py3-none-any.whl.metadata (110 kB)
Collecting pydantic-settings<3,>=2.5.2 (from eth-ape)
Downloading pydantic_settings-2.15.0-py3-none-any.whl.metadata (3.9 kB)
Collecting pytest<9.0,>=8.0 (from eth-ape)
Downloading pytest-8.4.2-py3-none-any.whl.metadata (7.7 kB)
Requirement already satisfied: python-dateutil<3,>=2.8.2 in d:\Study\IS355\.venv-ape\lib\site-packages (from eth-ape) (2.9.0.post0)
Collecting PyYAML<7,>=5.1 (from eth-ape)
Downloading pyyaml-6.0.3-cp312-cp312-win_amd64.whl.metadata (2.4 kB)
Collecting requests<3,>=2.28.1 (from eth-ape)
Using cached requests-2.34.2-py3-none-any.whl.metadata (4.8 kB)
Collecting rich<15,>=14.3.3 (from eth-ape)
Downloading rich-14.3.4-py3-none-any.whl.metadata (18 kB)
Collecting SQLAlchemy>=1.4.35 (from eth-ape)
Downloading sqlalchemy-2.1.3-cp312-cp312-win_amd64.whl.metadata (9.9 kB)
Collecting tqdm<5.0,>=4.67 (from eth-ape)
Downloading tqdm-4.70.1-py3-none-any.whl.metadata (57 kB)
Requirement already satisfied: traitlets>=5.3.0 in d:\Study\IS355\.venv-ape\lib\site-packages (from eth-ape) (5.16.1)
Collecting urllib3<3,>=2.3 (from eth-ape)
Using cached urllib3-2.8.0-py3-none-any.whl.metadata (7.4 kB)
Collecting watchdog<4,>=3.0 (from eth-ape)
Downloading watchdog-3.0.0-py3-none-win_amd64.whl.metadata (36 kB)
Collecting eth-abi<6,>=5.1.0 (from eth-ape)
Downloading eth_abi-5.2.0-py3-none-any.whl.metadata (3.8 kB)
Collecting eth-account<0.14,>=0.11.3 (from eth-ape)
Downloading eth_account-0.13.7-py3-none-any.whl.metadata (3.7 kB)

```

Collecting eth-typing<6,>=3.5.2 (from eth-ape)
  Downloading eth_typing-5.2.1-py3-none-any.whl.metadata (3.2 kB)
Collecting eth-utils<6,>=2.1.0 (from eth-ape)
  Downloading eth_utils-5.3.1-py3-none-any.whl.metadata (5.7 kB)
Collecting hexbytes<2,>=1.2.0 (from eth-ape)
  Downloading hexbytes-1.3.1-py3-none-any.whl.metadata (3.3 kB)
Collecting py-geth<7,>=6.2.0 (from eth-ape)
  Downloading py_geth-6.4.0-py3-none-any.whl.metadata (9.3 kB)
Collecting trie<4,>=3.0.1 (from eth-ape)
  Downloading trie-3.1.0-py3-none-any.whl.metadata (19 kB)
Collecting web3<8,>=7.12.1 (from web3[tester]<8,>=7.12.1->eth-ape)
  Downloading web3-7.16.0-py3-none-any.whl.metadata (5.6 kB)
Collecting ethpm-types<0.7,>=0.6.28 (from eth-ape)
  Downloading ethpm_types-0.6.32-py3-none-any.whl.metadata (3.2 kB)
Collecting eip712<0.4,>=0.3.0 (from eth-ape)
  Downloading eip712-0.3.3-py3-none-any.whl.metadata (3.4 kB)
Collecting eth_pydantic_types<0.3,>=0.2.1 (from eth-ape)
  Downloading eth_pydantic_types-0.2.7-py3-none-any.whl.metadata (6.2 kB)
Collecting evmchains<0.2,>=0.1.0 (from eth-ape)
  Downloading evmchains-0.1.8-py3-none-any.whl.metadata (2.6 kB)
Collecting evm-trace<0.4,>=0.3.0 (from eth-ape)
  Downloading evm_trace-0.3.0-py3-none-any.whl.metadata (7.7 kB)
Requirement already satisfied: six>=1.12.0 in d:\Study\IS355\venv-ape\Lib\site-pack
ages (from asttokens<3,>=2.4.1->eth-ape) (1.17.0)
Collecting eth-hash (from cchecksum<1,>=0.0.3->eth-ape)
  Using cached eth_hash-0.8.0-py3-none-any.whl.metadata (4.1 kB)
Collecting safe-pysha3>=1.0.0 (from cchecksum<1,>=0.0.3->eth-ape)
  Downloading safe_pysha3-1.0.5.tar.gz (827 kB)
----- 0.0/827.8 kB ? eta -:-:--
----- 262.1/827.8 kB ? eta -:-:--
----- 827.8/827.8 kB 3.6 MB/s 0:00:00
Installing build dependencies: started
Installing build dependencies: finished with status 'done'
Getting requirements to build wheel: started
Getting requirements to build wheel: finished with status 'done'
Preparing metadata (pyproject.toml): started
Preparing metadata (pyproject.toml): finished with status 'done'
Requirement already satisfied: colorama in d:\Study\IS355\venv-ape\Lib\site-package
s (from click<8.2,>=8.1.6->eth-ape) (0.4.6)
Collecting parsimonious<0.11.0,>=0.10.0 (from eth-abi<6,>=5.1.0->eth-ape)
  Using cached parsimonious-0.10.0-py3-none-any.whl.metadata (25 kB)
Collecting bitarray>=2.4.0 (from eth-account<0.14,>=0.11.3->eth-ape)
  Downloading bitarray-3.11.0-cp312-cp312-win_amd64.whl.metadata (39 kB)
Collecting eth-keyfile<0.9.0,>=0.7.0 (from eth-account<0.14,>=0.11.3->eth-ape)
  Downloading eth_keyfile-0.8.1-py3-none-any.whl.metadata (8.5 kB)
Collecting eth-keys>=0.4.0 (from eth-account<0.14,>=0.11.3->eth-ape)
  Using cached eth_keys-0.8.0-py3-none-any.whl.metadata (12 kB)
Collecting eth-rlp>=2.1.0 (from eth-account<0.14,>=0.11.3->eth-ape)
  Using cached eth_rlp-3.0.0-py3-none-any.whl.metadata (2.2 kB)
Collecting rlp>=1.0.0 (from eth-account<0.14,>=0.11.3->eth-ape)
  Using cached rlp-5.0.0-py3-none-any.whl.metadata (2.1 kB)
Collecting ckzg>=2.0.0 (from eth-account<0.14,>=0.11.3->eth-ape)
  Downloading ckzg-2.1.8-cp312-cp312-win_amd64.whl.metadata (930 bytes)
Collecting pycryptodome<4,>=3.6.6 (from eth-keyfile<0.9.0,>=0.7.0->eth-account<0.14,
>=0.11.3->eth-ape)
  Downloading pycryptodome-3.24.0-cp37-abi3-win_amd64.whl.metadata (3.5 kB)

```

Requirement already satisfied: typing_extensions<5,>=4.8.0 in d:\Study\IS355\.venv-ape\Lib\site-packages (from eth_pydantic_types<0.3,>=0.2.1->eth-ape) (4.16.0)

Collecting cytoolz>=0.10.1 (from eth-utils<6,>=2.1.0->eth-ape)

 Downloading cytoolz-1.1.0-cp312-cp312-win_amd64.whl.metadata (5.3 kB)

Collecting py-cid<0.4,>=0.3.0 (from ethpm-types<0.7,>=0.6.28->eth-ape)

 Downloading py_cid-0.3.1-py3-none-any.whl.metadata (4.4 kB)

Collecting msgspec>=0.8 (from evm-trace<0.4,>=0.3.0->eth-ape)

 Downloading msgspec-0.22.0-cp312-cp312-win_amd64.whl.metadata (6.1 kB)

Collecting py-evm<0.13,>=0.10.1b1 (from evm-trace<0.4,>=0.3.0->eth-ape)

 Downloading py_evm-0.12.1b1-py3-none-any.whl.metadata (6.1 kB)

Collecting decorator (from ipython<9,>=8.18.1->eth-ape)

 Using cached decorator-5.3.1-py3-none-any.whl.metadata (3.9 kB)

Requirement already satisfied: jedi>=0.16 in d:\Study\IS355\.venv-ape\Lib\site-packages (from ipython<9,>=8.18.1->eth-ape) (0.20.0)

Requirement already satisfied: matplotlib-inline in d:\Study\IS355\.venv-ape\Lib\site-packages (from ipython<9,>=8.18.1->eth-ape) (0.2.2)

Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in d:\Study\IS355\.venv-ape\Lib\site-packages (from ipython<9,>=8.18.1->eth-ape) (3.0.53)

Requirement already satisfied: pygments>=2.4.0 in d:\Study\IS355\.venv-ape\Lib\site-packages (from ipython<9,>=8.18.1->eth-ape) (2.21.0)

Requirement already satisfied: stack_data in d:\Study\IS355\.venv-ape\Lib\site-packages (from ipython<9,>=8.18.1->eth-ape) (0.6.3)

Collecting pytz>=2020.1 (from pandas<3,>=2.2.2->eth-ape)

 Downloading pytz-2026.5-py2.py3-none-any.whl.metadata (22 kB)

Collecting tzdata>=2022.7 (from pandas<3,>=2.2.2->eth-ape)

 Downloading tzdata-2026.5-py2.py3-none-any.whl.metadata (1.4 kB)

Collecting regex>=2022.3.15 (from parsimonious<0.11.0,>=0.10.0->eth-abi<6,>=5.1.0->eth-ape)

 Downloading regex-2026.9.29-cp312-cp312-win_amd64.whl.metadata (41 kB)

Requirement already satisfied: wcwidth>=0.1.4 in d:\Study\IS355\.venv-ape\Lib\site-packages (from prompt_toolkit<3.1.0,>=3.0.41->ipython<9,>=8.18.1->eth-ape) (0.9.2)

Collecting base58<2.0,>=1.0.2 (from py-cid<0.4,>=0.3.0->ethpm-types<0.7,>=0.6.28->eth-ape)

 Downloading base58-1.0.3-py3-none-any.whl.metadata (565 bytes)

Collecting py-multibase<2.0.0,>=1.0.0 (from py-cid<0.4,>=0.3.0->ethpm-types<0.7,>=0.6.28->eth-ape)

 Downloading py_multibase-1.0.3-py2.py3-none-any.whl.metadata (3.5 kB)

Collecting py-multicodec<0.3.0 (from py-cid<0.4,>=0.3.0->ethpm-types<0.7,>=0.6.28->eth-ape)

 Downloading py_multicodec-0.2.1-py2.py3-none-any.whl.metadata (3.2 kB)

Collecting morphys<2.0,>=1.0 (from py-cid<0.4,>=0.3.0->ethpm-types<0.7,>=0.6.28->eth-ape)

 Downloading morphys-1.0-py2.py3-none-any.whl.metadata (3.3 kB)

Collecting pymultihash<1.0.0,>=0.8.0 (from py-cid<0.4,>=0.3.0->ethpm-types<0.7,>=0.6.28->eth-ape)

 Downloading pymultihash-0.8.2-py3-none-any.whl.metadata (1.8 kB)

Collecting cached-property>=1.5.1 (from py-evm<0.13,>=0.10.1b1->evm-trace<0.4,>=0.3.0->eth-ape)

 Downloading cached_property-2.0.1-py3-none-any.whl.metadata (10 kB)

Collecting eth-bloom>=1.0.3 (from py-evm<0.13,>=0.10.1b1->evm-trace<0.4,>=0.3.0->eth-ape)

 Downloading eth_bloom-4.0.0-py3-none-any.whl.metadata (7.5 kB)

Collecting lru-dict>=1.1.6 (from py-evm<0.13,>=0.10.1b1->evm-trace<0.4,>=0.3.0->eth-ape)

 Downloading lru_dict-1.4.1-cp312-cp312-win_amd64.whl.metadata (4.9 kB)

Collecting py-ecc>=8.0.0 (from py-evm<0.13,>=0.10.1b1->evm-trace<0.4,>=0.3.0->eth-ape)

e)

Using cached py_ecc-8.0.0-py3-none-any.whl.metadata (3.1 kB)
Collecting semantic-version>=2.6.0 (from py-geth<7,>=6.2.0->eth-ape)
Downloading semantic_version-2.10.0-py2.py3-none-any.whl.metadata (9.7 kB)
Collecting types-requests>=2.0.0 (from py-geth<7,>=6.2.0->eth-ape)
Downloading types_requests-2.33.0.20260906-py3-none-any.whl.metadata (2.2 kB)
Collecting python-baseconv<2.0,>=1.2.0 (from py-multibase<2.0.0,>=1.0.0->py-cid<0.4,>=0.3.0->ethpm-types<0.7,>=0.6.28->eth-ape)
Downloading python-baseconv-1.2.2.tar.gz (4.9 kB)
Installing build dependencies: started
Installing build dependencies: finished with status 'done'
Getting requirements to build wheel: started
Getting requirements to build wheel: finished with status 'done'
Preparing metadata (pyproject.toml): started
Preparing metadata (pyproject.toml): finished with status 'done'
Collecting varint<2.0.0,>=1.0.2 (from py-multicodec<0.3.0->py-cid<0.4,>=0.3.0->ethpm-types<0.7,>=0.6.28->eth-ape)
Downloading varint-1.0.2.tar.gz (1.9 kB)
Installing build dependencies: started
Installing build dependencies: finished with status 'done'
Getting requirements to build wheel: started
Getting requirements to build wheel: finished with status 'done'
Preparing metadata (pyproject.toml): started
Preparing metadata (pyproject.toml): finished with status 'done'
Collecting annotated-types>=0.6.0 (from pydantic<3,>=2.10->eth-ape)
Using cached annotated_types-0.8.0-py3-none-any.whl.metadata (15 kB)
Collecting pydantic-core==2.46.5 (from pydantic<3,>=2.10->eth-ape)
Downloading pydantic_core-2.46.5-cp312-cp312-win_amd64.whl.metadata (6.7 kB)
Collecting typing-inspection>=0.4.2 (from pydantic<3,>=2.10->eth-ape)
Using cached typing_inspection-0.4.4-py3-none-any.whl.metadata (2.6 kB)
Collecting python-dotenv>=0.21.0 (from pydantic-settings<3,>=2.5.2->eth-ape)
Downloading python_dotenv-1.2.4-py3-none-any.whl.metadata (29 kB)
Collecting iniconfig>=1 (from pytest<9.0,>=8.0->eth-ape)
Downloading iniconfig-2.3.0-py3-none-any.whl.metadata (2.5 kB)
Collecting charset_normalizer<4,>=2 (from requests<3,>=2.28.1->eth-ape)
Downloading charset_normalizer-3.5.2-cp312-cp312-win_amd64.whl.metadata (47 kB)
Collecting idna<4,>=2.5 (from requests<3,>=2.28.1->eth-ape)
Using cached idna-3.20-py3-none-any.whl.metadata (7.2 kB)
Collecting certifi>=2023.5.7 (from requests<3,>=2.28.1->eth-ape)
Using cached certifi-2026.7.22-py3-none-any.whl.metadata (2.5 kB)
Collecting markdown-it-py>=2.2.0 (from rich<15,>=14.3.3->eth-ape)
Downloading markdown_it_py-4.2.0-py3-none-any.whl.metadata (7.4 kB)
Collecting sortedcontainers>=2.1.0 (from trie<4,>=3.0.1->eth-ape)
Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl.metadata (10 kB)
Collecting aiohttp>=3.7.4.post0 (from web3<8,>=7.12.1->web3[tester]<8,>=7.12.1->eth-ape)
Downloading aiohttp-3.14.4-cp312-cp312-win_amd64.whl.metadata (8.5 kB)
Collecting pywin32>=223 (from web3<8,>=7.12.1->web3[tester]<8,>=7.12.1->eth-ape)
Downloading pywin32-312-cp312-cp312-win_amd64.whl.metadata (11 kB)
Collecting websockets<16.0.0,>=10.0.0 (from web3<8,>=7.12.1->web3[tester]<8,>=7.12.1->eth-ape)
Downloading websockets-15.0.1-cp312-cp312-win_amd64.whl.metadata (7.0 kB)
Collecting pyunormalize>=15.0.0 (from web3<8,>=7.12.1->web3[tester]<8,>=7.12.1->eth-ape)
Using cached pyunormalize-18.0.0-py3-none-any.whl.metadata (5.5 kB)
Collecting eth-tester<0.14.0b1,>=0.13.0b1 (from eth-tester[py-evm]<0.14.0b1,>=0.13.0

```

b1; extra == "tester"->web3[tester]<8,>=7.12.1->eth-ape)
  Downloading eth_tester-0.13.0b1-py3-none-any.whl.metadata (38 kB)
Collecting aiohappyeyeballs>=2.5.0 (from aiohttp>=3.7.4.post0->web3<8,>=7.12.1->web3
[tester]<8,>=7.12.1->eth-ape)
  Using cached aiohappyeyeballs-2.7.1-py3-none-any.whl.metadata (5.9 kB)
Collecting aiosignal>=1.4.0 (from aiohttp>=3.7.4.post0->web3<8,>=7.12.1->web3[teste
r]<8,>=7.12.1->eth-ape)
  Using cached aiosignal-1.4.0-py3-none-any.whl.metadata (3.7 kB)
Collecting attrs>=17.3.0 (from aiohttp>=3.7.4.post0->web3<8,>=7.12.1->web3[tester]<
8,>=7.12.1->eth-ape)
  Using cached attrs-26.1.0-py3-none-any.whl.metadata (8.8 kB)
Collecting frozenlist>=1.1.1 (from aiohttp>=3.7.4.post0->web3<8,>=7.12.1->web3[teste
r]<8,>=7.12.1->eth-ape)
  Downloading frozenlist-1.8.0-cp312-cp312-win_amd64.whl.metadata (21 kB)
Collecting multidict<7.0,>=4.5 (from aiohttp>=3.7.4.post0->web3<8,>=7.12.1->web3[tes
ter]<8,>=7.12.1->eth-ape)
  Downloading multidict-6.9.1-cp312-cp312-win_amd64.whl.metadata (5.4 kB)
Collecting propcache>=0.2.0 (from aiohttp>=3.7.4.post0->web3<8,>=7.12.1->web3[teste
r]<8,>=7.12.1->eth-ape)
  Downloading propcache-0.5.4-cp312-cp312-win_amd64.whl.metadata (25 kB)
Collecting yarl<2.0,>=1.25.1 (from aiohttp>=3.7.4.post0->web3<8,>=7.12.1->web3[teste
r]<8,>=7.12.1->eth-ape)
  Downloading yarl-1.25.1-cp312-cp312-win_amd64.whl.metadata (115 kB)
Collecting toolz>=0.8.0 (from cytoolz>=0.10.1->eth-utils<6,>=2.1.0->eth-ape)
  Using cached toolz-1.1.0-py3-none-any.whl.metadata (5.1 kB)
Requirement already satisfied: parso<0.9.0,>=0.8.6 in d:\Study\IS355\.venv-ape\Lib\s
ite-packages (from jedi>=0.16->ipython<9,>=8.18.1->eth-ape) (0.8.7)
Collecting mdurl~=0.1 (from markdown-it-py>=2.2.0->rich<15,>=14.3.3->eth-ape)
  Downloading mdurl-0.1.2-py3-none-any.whl.metadata (1.6 kB)
Requirement already satisfied: executing>=1.2.0 in d:\Study\IS355\.venv-ape\Lib\site
-packages (from stack_data->ipython<9,>=8.18.1->eth-ape) (2.2.1)
Requirement already satisfied: pure-eval in d:\Study\IS355\.venv-ape\Lib\site-packag
es (from stack_data->ipython<9,>=8.18.1->eth-ape) (0.2.4)
Downloading eth_ape-0.8.52-py3-none-any.whl (430 kB)
Downloading asttokens-2.4.1-py2.py3-none-any.whl (27 kB)
Downloading cchecksum-0.4.5-cp312-cp312-win_amd64.whl (252 kB)
Downloading click-8.1.8-py3-none-any.whl (98 kB)
Downloading eip712-0.3.3-py3-none-any.whl (11 kB)
Downloading eth_abi-5.2.0-py3-none-any.whl (28 kB)
Downloading eth_account-0.13.7-py3-none-any.whl (587 kB)
----- 0.0/587.5 kB ? eta -:-:--
----- 587.5/587.5 kB 6.9 MB/s 0:00:00
Downloading eth_keyfile-0.8.1-py3-none-any.whl (7.5 kB)
Downloading eth_pydantic_types-0.2.7-py3-none-any.whl (19 kB)
Downloading eth_typing-5.2.1-py3-none-any.whl (19 kB)
Downloading eth_utils-5.3.1-py3-none-any.whl (102 kB)
Downloading ethpm_types-0.6.32-py3-none-any.whl (35 kB)
Downloading evm_trace-0.3.0-py3-none-any.whl (23 kB)
Downloading evmchains-0.1.8-py3-none-any.whl (18 kB)
Downloading hexbytes-1.3.1-py3-none-any.whl (5.1 kB)
Downloading ijson-3.5.1-cp312-cp312-win_amd64.whl (54 kB)
Downloading ipython-8.39.0-py3-none-any.whl (831 kB)
----- 0.0/831.8 kB ? eta -:-:--
----- 831.8/831.8 kB 9.2 MB/s 0:00:00
Downloading numpy-1.26.4-cp312-cp312-win_amd64.whl (15.5 MB)
----- 0.0/15.5 MB ? eta -:-:--

```

```

----- 1.8/15.5 MB 8.4 MB/s eta 0:00:02
----- 3.4/15.5 MB 8.4 MB/s eta 0:00:02
----- 5.5/15.5 MB 8.8 MB/s eta 0:00:02
----- 7.9/15.5 MB 9.4 MB/s eta 0:00:01
----- 10.0/15.5 MB 9.7 MB/s eta 0:00:01
----- 12.6/15.5 MB 10.2 MB/s eta 0:00:01
----- 15.2/15.5 MB 10.5 MB/s eta 0:00:01
----- 15.5/15.5 MB 10.4 MB/s 0:00:01
Downloading packaging-23.2-py3-none-any.whl (53 kB)
Downloading pandas-2.3.3-cp312-cp312-win_amd64.whl (11.0 MB)
----- 0.0/11.0 MB ? eta -:-:--
----- 3.4/11.0 MB 16.7 MB/s eta 0:00:01
----- 6.3/11.0 MB 15.4 MB/s eta 0:00:01
----- 10.2/11.0 MB 16.8 MB/s eta 0:00:01
----- 11.0/11.0 MB 16.0 MB/s 0:00:00
Using cached parsimonious-0.10.0-py3-none-any.whl (48 kB)
Downloading pluggy-1.6.0-py3-none-any.whl (20 kB)
Downloading py_cid-0.3.1-py3-none-any.whl (6.5 kB)
Downloading base58-1.0.3-py3-none-any.whl (3.1 kB)
Downloading morphys-1.0-py2.py3-none-any.whl (5.6 kB)
Downloading py_evm-0.12.1b1-py3-none-any.whl (798 kB)
----- 0.0/798.6 kB ? eta -:-:--
----- 798.6/798.6 kB 16.8 MB/s 0:00:00
Downloading py_geth-6.4.0-py3-none-any.whl (28 kB)
Downloading py_multibase-1.0.3-py2.py3-none-any.whl (6.4 kB)
Downloading py_multicodec-0.2.1-py2.py3-none-any.whl (8.5 kB)
Downloading pycryptodome-3.24.0-cp37-abi3-win_amd64.whl (1.8 MB)
----- 0.0/1.8 MB ? eta -:-:--
----- 1.8/1.8 MB 25.2 MB/s 0:00:00
Using cached pydantic-2.13.5-py3-none-any.whl (472 kB)
Downloading pydantic_core-2.46.5-cp312-cp312-win_amd64.whl (2.0 MB)
----- 0.0/2.0 MB ? eta -:-:--
----- 2.0/2.0 MB 22.8 MB/s 0:00:00
Downloading pydantic_settings-2.15.0-py3-none-any.whl (69 kB)
Downloading pymultihash-0.8.2-py3-none-any.whl (13 kB)
Downloading pytest-8.4.2-py3-none-any.whl (365 kB)
Downloading pyyaml-6.0.3-cp312-cp312-win_amd64.whl (154 kB)
Using cached requests-2.34.2-py3-none-any.whl (73 kB)
Downloading charset_normalizer-3.5.2-cp312-cp312-win_amd64.whl (207 kB)
Using cached idna-3.20-py3-none-any.whl (69 kB)
Downloading rich-14.3.4-py3-none-any.whl (310 kB)
Downloading tqdm-4.70.1-py3-none-any.whl (80 kB)
Downloading trie-3.1.0-py3-none-any.whl (38 kB)
Using cached urllib3-2.8.0-py3-none-any.whl (135 kB)
Downloading watchdog-3.0.0-py3-none-win_amd64.whl (82 kB)
Downloading web3-7.16.0-py3-none-any.whl (1.4 MB)
----- 0.0/1.4 MB ? eta -:-:--
----- 1.4/1.4 MB 17.9 MB/s 0:00:00
Downloading eth_tester-0.13.0b1-py3-none-any.whl (76 kB)
Using cached eth_hash-0.8.0-py3-none-any.whl (8.0 kB)
Downloading websockets-15.0.1-cp312-cp312-win_amd64.whl (176 kB)
Downloading aiohttp-3.14.4-cp312-cp312-win_amd64.whl (506 kB)
Downloading multidict-6.9.1-cp312-cp312-win_amd64.whl (59 kB)
Downloading yarl-1.25.1-cp312-cp312-win_amd64.whl (102 kB)
Using cached aiohappyeyeballs-2.7.1-py3-none-any.whl (15 kB)
Using cached aiosignal-1.4.0-py3-none-any.whl (7.5 kB)

```

```

Using cached annotated_types-0.8.0-py3-none-any.whl (13 kB)
Using cached attrs-26.1.0-py3-none-any.whl (67 kB)
Downloading bitarray-3.11.0-cp312-cp312-win_amd64.whl (171 kB)
Downloading cached_property-2.0.1-py3-none-any.whl (7.4 kB)
Using cached certifi-2026.7.22-py3-none-any.whl (136 kB)
Downloading ckzg-2.1.8-cp312-cp312-win_amd64.whl (103 kB)
Downloading cytoolz-1.1.0-cp312-cp312-win_amd64.whl (947 kB)
----- 0.0/948.0 kB ? eta -:-:--
----- 948.0/948.0 kB 14.6 MB/s 0:00:00
Downloading eth_bloom-4.0.0-py3-none-any.whl (5.5 kB)
Using cached eth_keys-0.8.0-py3-none-any.whl (20 kB)
Using cached eth_rlp-3.0.0-py3-none-any.whl (15 kB)
Downloading frozenlist-1.8.0-cp312-cp312-win_amd64.whl (44 kB)
Downloading iniconfig-2.3.0-py3-none-any.whl (7.5 kB)
Downloading lru_dict-1.4.1-cp312-cp312-win_amd64.whl (14 kB)
Downloading markdown_it_py-4.2.0-py3-none-any.whl (91 kB)
Downloading mdurl-0.1.2-py3-none-any.whl (10.0 kB)
Downloading msgspec-0.22.0-cp312-cp312-win_amd64.whl (192 kB)
Downloading propcache-0.5.4-cp312-cp312-win_amd64.whl (46 kB)
Using cached py_ecc-8.0.0-py3-none-any.whl (47 kB)
Downloading python_dotenv-1.2.4-py3-none-any.whl (23 kB)
Downloading pytz-2026.5-py2.py3-none-any.whl (506 kB)
Using cached pyunormalize-18.0.0-py3-none-any.whl (78 kB)
Downloading pywin32-312-cp312-cp312-win_amd64.whl (6.9 MB)
----- 0.0/6.9 MB ? eta -:-:--
----- 4.5/6.9 MB 20.7 MB/s eta 0:00:01
----- 6.9/6.9 MB 18.5 MB/s 0:00:00
Downloading regex-2026.9.29-cp312-cp312-win_amd64.whl (280 kB)
Using cached rlp-5.0.0-py3-none-any.whl (19 kB)
Downloading semantic_version-2.10.0-py2.py3-none-any.whl (15 kB)
Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl (29 kB)
Downloading sqlalchemy-2.1.3-cp312-cp312-win_amd64.whl (2.4 MB)
----- 0.0/2.4 MB ? eta -:-:--
----- 2.4/2.4 MB 23.3 MB/s 0:00:00
Using cached toolz-1.1.0-py3-none-any.whl (58 kB)
Downloading types_requests-2.33.0.20260906-py3-none-any.whl (21 kB)
Using cached typing_inspection-0.4.4-py3-none-any.whl (14 kB)
Downloading tzdata-2026.5-py2.py3-none-any.whl (347 kB)
Using cached decorator-5.3.1-py3-none-any.whl (10 kB)
Building wheels for collected packages: python-baseconv, varint, lazyasd, safe-pysha
3
  Building wheel for python-baseconv (pyproject.toml): started
  Building wheel for python-baseconv (pyproject.toml): finished with status 'done'
  Created wheel for python-baseconv: filename=python_baseconv-1.2.2-py3-none-any.whl
size=5592 sha256=1817f57dbede56e0dc6a80ad5c9b9642ca942b87b51cdaeaca58e59d7cdcb737
  Stored in directory: c:\users\pc\appdata\local\pip\cache\wheels\8a\78\af\94a946893
407538393c958b958f76c68527ce2aaa9ff156df7
  Building wheel for varint (pyproject.toml): started
  Building wheel for varint (pyproject.toml): finished with status 'done'
  Created wheel for varint: filename=varint-1.0.2-py3-none-any.whl size=2017 sha256=
915faa7f8d3c07a54a5830a567e855c5ca9889947520f01b98e559d9bf1414ce
  Stored in directory: c:\users\pc\appdata\local\pip\cache\wheels\9c\81\f4\3d93402a4
7e78b0e782062a5819b07b424427af0c4ed6655ab
  Building wheel for lazyasd (pyproject.toml): started
  Building wheel for lazyasd (pyproject.toml): finished with status 'done'
  Created wheel for lazyasd: filename=lazyasd-0.1.4-py3-none-any.whl size=17618 sha2

```

```

56=e08abadcb11b6334886850fff7f3211fc58f0316e8d8ea72d97911206cca6fc8
  Stored in directory: c:\users\pc\appdata\local\pip\cache\wheels\1f\2b\48\3a5f159c4
be8add66ffc4f0f9b3db2a4e60c1202dc8f0b9b23
  Building wheel for safe-pysha3 (pyproject.toml): started
  Building wheel for safe-pysha3 (pyproject.toml): finished with status 'error'
Successfully built python-baseconv varint lazyasd
Failed to build safe-pysha3
Note: you may need to restart the kernel to use updated packages.
      error: subprocess-exited-with-error

      × Building wheel for safe-pysha3 (pyproject.toml) did not run successfully.
        | exit code: 1
        |↳ [22 lines of output]
          C:\Users\PC\AppData\Local\Temp\pip-build-env-a2u301f5\overlay\Lib\site-package
s\setuptools\dist.py:765: SetuptoolsDeprecationWarning: License classifiers are depr
ecated.
          !!

          *****
          *****

          Please consider removing the following classifiers in favor of a SPDX
license expression:

          License :: OSI Approved :: Python Software Foundation License
          License :: CC0 1.0 Universal (CC0 1.0) Public Domain Dedication

          See https://packaging.python.org/en/latest/guides/writing-pyproject-to
ml/#license for details.
          *****
          *****

          !!
          self._finalize_license_expression()
          running bdist_wheel
          running build
          running build_py
          creating build\lib.win-amd64-cpython-312
          copying sha3.py -> build\lib.win-amd64-cpython-312
          running build_ext
          building '_pysha3' extension
          error: Microsoft Visual C++ 14.0 or greater is required. Get it with "Microsof
t C++ Build Tools": https://visualstudio.microsoft.com/visual-cpp-build-tools/
          [end of output]

          note: This error originates from a subprocess, and is likely not a problem with pi
p.
          ERROR: Failed building wheel for safe-pysha3
          error: failed-wheel-build-for-install

      × Failed to build installable wheels for some pyproject.toml based projects
        |↳ safe-pysha3

```

Bước 2 — Kiểm tra version

In [12]: `!ape --version`

'ape' is not recognized as an internal or external command,
operable program or batch file.

Bước 3 — Tạo project (ape init)

```
In [13]: # Tao project Ape (co the hoi ten project – nhap Bai02)
# Neu dang o TH/w2 va da co san contracts/scripts thi bo qua buoc nay
!ape init
```

'ape' is not recognized as an internal or external command,
operable program or batch file.

Bước 4 — Liệt kê cấu trúc thư mục sau ape init

```
In [14]: from pathlib import Path

root = Path(".")
print("CWD:", root.resolve())
print()
for p in sorted(root.iterdir()):
    kind = "dir " if p.is_dir() else "file"
    print(f" [{kind}] {p.name}")

# Cac thu muc thuong gap
for name in ["contracts", "scripts", "tests", "ape-config.yaml", ".build"]:
    print(f" exists {name}? -> {(root / name).exists()}")
```

CWD: D:\Study\IS355\TH\w2

```
[dir ] _pdf_imgs
[file] Bai02.pdf
[file] BT2_TrenLop.ipynb
[dir ] ganache
[dir ] Images
exists contracts? -> False
exists scripts? -> False
exists tests? -> False
exists ape-config.yaml? -> False
exists .build? -> False
```

Bước 5 — Plugin Vyper

```
In [15]: !ape plugins list
```

'ape' is not recognized as an internal or external command,
operable program or batch file.

```
In [16]: # Cai plugin Vyper neu chua co
!ape plugins install vyper
```

'ape' is not recognized as an internal or external command,
operable program or batch file.

```
In [17]: # Kiem tra lai
!ape plugins list
```

'ape' is not recognized as an internal or external command,
operable program or batch file.

ii. Contract SimpleStorage.vy

```
In [5]: # Doc file contracts/SimpleStorage.vy (da tach rieng)
        from pathlib import Path

        p = Path("contracts/SimpleStorage.vy")
        assert p.exists(), f"Chua co {p.resolve()} -kiem tra thu muc TH/w2"
        print(f"File: {p.resolve()}\n")
        print(p.read_text(encoding="utf-8"))
```

File: D:\Study\IS355\TH\w2\contracts\SimpleStorage.vy

```
# @version ^0.4.3
```

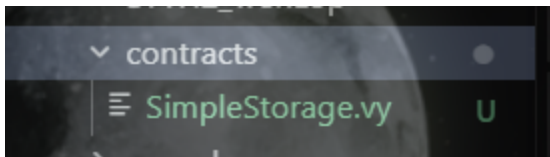
```
num: uint256
```

```
@external
def store(num: uint256):
    self.num = num
```

```
@external
def retrieve() -> uint256:
    return self.num
```

File đã tách tại contracts/SimpleStorage.vy :

Explorer:



Nội dung file:

```
BT2_TrenLop.ipynb U ● SimpleStorage.vy U X
TH > w2 > contracts > SimpleStorage.vy
1  # @version ^0.4.3
2
3  num: uint256
4
5  @external
6  def store(num: uint256):
7      self.num = num
8
9  @external
10 def retrieve() -> uint256:
11     return self.num
12
```

Biên dịch contract — `ape compile`

```
In [7]: # Chạy trong TH/w2 (có contracts/SimpleStorage.vy)
!ape compile
```

SUCCESS: 'local project' compiled.

D:\Study\IS355\.venv-ape\Lib\site-packages\pydantic_settings\sources\utils.py:47: IncompleteFieldDefinitionWarning: Field 'import_remapping' has an incomplete definition: its annotation contains an unresolved forward reference, so settings sources may fail to correctly resolve its value. Call `model\_rebuild()` on the model where the field is defined, once all the referenced types are defined.
warnings.warn(

iii. Account alias trên Ganache → lỗi `invalid opcode`

Chạy ở **terminal**.

Lệnh (PowerShell)

```
In [ ]: cd D:\Study\IS355\TH\w2
D:\Study\IS355\.venv-ape\Scripts\ape.exe accounts import is355_
```

Private key **INDEX 4**

Address: `0x4331381a3254dF46ff30D0CFF70c78EE159398F9`

Khi `ape accounts import is355_` hỏi, nhập lần lượt:

1. **Enter Private Key:** dán key trong ảnh trên
2. **Create Passphrase:** 1234567890
3. **Repeat for confirmation:** 1234567890

Kết quả import trên máy (address INDEX 4):

```
PS D:\Study\IS355\TH\w2> D:\Study\IS355\.venv-ape\Scripts\ape.exe accounts import is355_
Enter Private Key:
Create Passphrase to encrypt account:
Repeat for confirmation:
SUCCESS: A new account '0x4331381a3254dF46ff30D0CFF70c78EE159398F9' has been added with the id 'is355_'
```

Alias đang dùng: `is355_` — trong `deploy.py` gọi `accounts.load("is355_")`.

Kiểm tra:

In []: `D:\Study\IS355\.venv-ape\Scripts\ape.exe accounts list`

```
PS D:\Study\IS355\TH\w2> D:\Study\IS355\.venv-ape\Scripts\ape.exe accounts list
Found 1 account:
0x4331381a3254dF46ff30D0CFF70c78EE159398F9 (alias: 'is355_')
```

Alias: `is355_` — address INDEX 4.

Sau khi import, nếu **deploy/gọi contract trên Ganache** có thể gặp:

```
VM Exception while processing transaction: invalid opcode
```

Giải thích ngắn (ghi vào báo cáo):

- Ganache (hardfork / EVM) đôi khi **không khớp** bytecode Vyper 0.4.x (opcode mới).
- Cách xử lý trong lab: chuyển sang **Geth --dev** để deploy / test (phần C).

```
In [ ]: # Chỉ copy lệnh này sang PowerShell (dùng chạy ape accounts import bang ! o day)
print(r'''
cd D:\Study\IS355\TH\w2
D:\Study\IS355\.venv-ape\Scripts\Activate.ps1
ape accounts import is355_

# Sau đó:
# 1) dán private key INDEX 4 (Ganache)
# 2) đặt passphrase (vd 1234567890)
# 3) ape accounts list
''')
```

C — Geth (Windows)

Phần này dùng Geth thay Ganache để deploy/test Ape ổn định hơn.

Không kèm ảnh mẫu đề — tự chạy rồi chụp kết quả máy bạn.

Dùng CLI: `D:\Study\IS355\.venv-ape\Scripts\ape.exe` (Python 3.12).

i. Cài Geth & chạy node `--dev`

Cách làm

1. Geth đã cài tại `D:\Tools\Geth\geth.exe` (v1.17.7).
2. Kiểm tra version:

`D:\Tools\Geth\geth.exe version`

```
PS D:\Study\IS355> D:\Tools\Geth\geth.exe version
Geth
Version: 1.17.7-stable
Git Commit: 3d858f858a458effb2a563788aedef1fe65e1f0d3
Git Commit Date: 20260930
Architecture: amd64
Go Version: go1.27.1
Operating System: windows
GOPATH=
GOROOT=
```

3. Chạy node (**giữ cửa sổ này mở**). Nếu lỗi IPC pipe thì thêm `--ipcdisable` :

`D:\Tools\Geth\geth.exe --dev --http --http.api eth,web3,net --http.addr 127.0.0.1 --http.port 8545 --ipcdisable`

```
PS D:\Study\IS355> D:\Tools\Geth\geth.exe --dev --http --http.api eth,web3,net --http.addr 127.0.0.1 --http.port 8545
INFO [10-06|10:50:54.091] Starting Geth on Ethereum mainnet...
INFO [10-06|10:50:54.094] Maximum peer count                      ETH=50 total=50
INFO [10-06|10:50:54.099] Set global gas cap                      cap=50,000,000
INFO [10-06|10:50:54.099] Engine API maximum reorg depth          depth=32
INFO [10-06|10:50:54.235] Using developer account                  address=0x71562b71999873DB5b286dF957af199Ec94617F7
INFO [10-06|10:50:54.235] Initializing the KZG library              backend=gokzg
INFO [10-06|10:50:54.236] Allocated trie memory caches             clean=614.00MiB dirty=1.00GiB
INFO [10-06|10:50:54.236] State schema set to default              scheme=path
WARN [10-06|10:50:54.236] Head block is not reachable
INFO [10-06|10:50:54.236] Initialising Ethereum protocol           network=1337 dbversion=<nil>
WARN [10-06|10:50:54.236] Sanitizing invalid node buffer size       provided=1.00GiB updated=256.00MiB
INFO [10-06|10:50:54.237] Load database journal from disk
INFO [10-06|10:50:54.240] State snapshot generator is not found
INFO [10-06|10:50:54.240] Starting snapshot generation              root=56e81f..63b421 accounts=0 slots=0 storage=0.00B
dangling=0 elapsed=0s
INFO [10-06|10:50:54.240] Initialized path database                 triecache=614.00MiB statecache=409.00MiB buffer=256.0
0MiB state-history="last 90000 blocks"
INFO [10-06|10:50:54.240] Writing custom genesis block
INFO [10-06|10:50:54.244] Resuming snapshot generation              root=56e81f..63b421 accounts=0 slots=0 storage=0.00B
dangling=0 elapsed=4.005ms
INFO [10-06|10:50:54.247] Generated snapshot                        accounts=0 slots=0 storage=0.00B dangling=0 elapsed=7
```

Network ID `1337` , HTTP port **8545**, account dev:

`0x71562b71999873DB5b286dF957af199Ec94617F7` .

Attach console (terminal thứ 2 — đừng tắt cửa sổ Geth)

```
D:\Tools\Geth\geth.exe attach http://127.0.0.1:8545
```

```
PS D:\Study\IS355> D:\Tools\Geth\geth.exe attach http://127.0.0.1:8545
Welcome to the Geth JavaScript console!

instance: Geth/v1.17.7-stable-3d858f85/windows-amd64/go1.27.1
at block: 0 (Thu Jan 01 1970 07:00:00 GMT+0700 (+07))
modules: eth:1.0 net:1.0 rpc:1.0 web3:1.0

To exit, press ctrl-d or type exit
```

Trong console Geth (prompt `>`), chạy:

```
eth.accounts
web3.fromWei(eth.getBalance(eth.accounts[0]), "ether")
Chụp địa chỉ eth.accounts[0] + số dư.
```

```
> eth.accounts
["0x71562b71999873db5b286df957af199ec94617f7"]
> web3.fromWei(eth.getBalance(eth.accounts[0]), "ether")
1.15792089237316195423570985008687907853269984665640564039457584007913129639927e+59
>
```

`--dev` cấp sẵn balance rất lớn cho `account[0]` — bình thường.

ii. Tạo account (`clef`) + chuyển ETH

Cách làm

```
cd D:\Study\IS355\TH\w2
mkdir my_keystore
clef newaccount --keystore my_keystore
```

- Đặt password và **nhớ**.
- Ghi báo cáo: địa chỉ mới + tên file `UTC--...` trong `my_keystore/`.

Trong `geth attach` (thay `0xDIA_CHI_MOI`):

```
eth.sendTransaction({
  from: eth.accounts[0],
  to: "0xDIA_CHI_MOI",
  value: web3.toWei(10, "ether")
})
web3.fromWei(eth.getBalance("0xDIA_CHI_MOI"), "ether")
Chụp: tạo account + send + balance.
```

iii. Script `send_eth_on_geth.py`

Cách làm

1. Chạy cell dưới → tạo file `send_eth_on_geth.py` (HTTP Windows, không IPC).
2. Sửa `password` / keystore nếu khác máy bạn.
3. Chạy (Geth đang mở):

```
cd D:\Study\IS355\TH\w2
```

```
D:\Study\IS355\.venv-ape\Scripts\python.exe send_eth_on_geth.py
```

4. Báo cáo: số ETH chuyển + phí (`gasUsed * gasPrice`). Chụp output.

Kết quả chạy (2 ETH, gasUsed=21000):

```
PS D:\Study\IS355\TH\w2> D:\Study\IS355\.venv-ape\Scripts\python.exe send_eth_on_geth.py
keystore: UTC--2026-10-06T04-03-59.286226600Z--69f1aeba4bf643a4c9533a6fd5eb64c8fc78fd5f
PRIVATE KEY (hex): 0x4a4f1b375a14008244ebd9dda0466dff43a4b4d712f5b1fc52dd175814f59206
sender   : 0x69f1AebA4Bf643A4C9533a6Fd5Eb64c8FC78fD5f
recipient: 0x71562b71999873DB5b286dF957af199Ec94617F7
tx hash  : 680ff1a1f5fbc0f6cc40ed9c2e7aee6d5ee510c5e79c8d18c8ad86c74c47c662
gasUsed  : 21000
value    : 2000000000000000000 wei = 2 ETH
phi ~    : 1050000000000000000 wei
```

```
In [ ]: from pathlib import Path
out = Path('send_eth_on_geth.py')
out.parent.mkdir(parents=True, exist_ok=True)
out.write_text('""""Chuyen ETH tren Geth (Windows: HTTP).""""\nfrom pathlib import Pa
print('Wrote', out.resolve())
print('Chay: D:\\Study\\IS355\\.venv-ape\\Scripts\\python.exe send_eth_on_geth.py')
```

(Tuỳ chọn) Import alias Ape từ key Geth

Account Ganache `is355_` **không có ETH trên Geth**. Muốn deploy bằng account keystore:

```
cd D:\Study\IS355\TH\w2
```

```
D:\Study\IS355\.venv-ape\Scripts\ape.exe accounts delete is355_ --yes
```

```
D:\Study\IS355\.venv-ape\Scripts\ape.exe accounts import is355_
```

Dán private key script vừa in + passphrase `1234567890` .

Hoặc fund ETH từ `eth.accounts[0]` sang địa chỉ Ganache rồi giữ alias cũ.

iv. Deploy SimpleStorage trên Geth bằng Ape

Cách làm

1. Cell dưới tạo `scripts/deploy.py` .
2. Kiểm tra: `contracts/SimpleStorage.vy` có sẵn, Geth chạy `:8545` , alias `is355_` có ETH.
3. Chạy:

```
cd D:\Study\IS355\TH\w2
D:\Study\IS355\.venv-ape\Scripts\ape.exe compile
D:\Study\IS355\.venv-ape\Scripts\ape.exe run deploy --network
ethereum:local:http://127.0.0.1:8545
```

4. Chụp: địa chỉ contract + Stored / Retrieved.

Compile OK:

```
PS D:\Study\IS355\TH\w2> D:\Study\IS355\.venv-ape\Scripts\ape.exe compile
D:\Study\IS355\.venv-ape\Lib\site-packages\pydantic_settings\sources\utils.py:47: IncompleteFieldDefinitionWarning: Field 'import_remapping' has an incomplete definition: its annotation contains an unresolved forward reference, so settings sources may fail to correctly resolve its value. Call `model_rebuild()` on the model where the field is defined, once all the referenced types are defined.
  warnings.warn(
SUCCESS: 'local project' compiled.
```

Deploy OK (Stored/Retrieved = 42):

```
PS D:\Study\IS355\TH\w2> D:\Study\IS355\.venv-ape\Scripts\ape.exe run deploy --network ethereum:local:http://127.0.0.1:8545
INFO: Connecting to existing Geth node at http://127.0.0.1:8545.
WARNING: Danger! This account will now sign any transaction it's given.
D:\Study\IS355\.venv-ape\Lib\site-packages\pydantic_settings\sources\utils.py:47: IncompleteFieldDefinitionWarning: Field 'import_remapping' has an incomplete definition: its annotation contains an unresolved forward reference, so settings sources may fail to correctly resolve its value. Call `model_rebuild()` on the model where the field is defined, once all the referenced types are defined.
  warnings.warn(
INFO: Contract will be deployed at: 0x57dcb76B638Ea754EFE65Ca891CDfA1b6b3250A0
WARNING: Using cached key for is355_
INFO: Confirmed 0xd72c81b4b7730ce40504f1d1d1efe14038eef9c43c457f935189badb5ef2b9f0 (total fees paid = 40177453103522)
)
SUCCESS: Contract 'SimpleStorage' deployed to: 0x57dcb76B638Ea754EFE65Ca891CDfA1b6b3250A0
Deployed at 0x57dcb76B638Ea754EFE65Ca891CDfA1b6b3250A0
WARNING: Using cached key for is355_
INFO: Confirmed 0xc1d432a3bf73596ec24fb65bf1e9d5c0dc9c6e27604edbbd5976f287b22a1a378 (total fees paid = 67559804705584)
)
Stored value 42
Retrieved value = 42
```

```
In [ ]: from pathlib import Path
out = Path('scripts/deploy.py')
out.parent.mkdir(parents=True, exist_ok=True)
out.write_text('from ape import accounts, project\n\n\ndef main():\n    dev = accounts[0]\n    print("Wrote", out.resolve())\n    print("Chạy: ape run deploy --network ethereum:local:http://127.0.0.1:8545")')
```

```
In [ ]: print("PowerShell:")
print(r" cd D:\Study\IS355\TH\w2")
print(r" D:\Study\IS355\.venv-ape\Scripts\ape.exe compile")
print(r" D:\Study\IS355\.venv-ape\Scripts\ape.exe run deploy --network ethereum:local:http://127.0.0.1:8545")')
```

v. Test tests/test_simple_storage.py

Cách làm

1. Cell dưới tạo file test.
2. Chạy:

```
cd D:\Study\IS355\TH\w2
D:\Study\IS355\.venv-ape\Scripts\ape.exe test
```

3. Chụp kết quả PASSED.

Giải thích file test

Thành phần	Vai trò
deployer	accounts[0] của môi trường test Ape
contract	Deploy mới SimpleStorage mỗi test
test_retrieve	Ban đầu retrieve() == 0
test_store	store(19) rồi retrieve() == 19
.call()	Đọc state, không tốn gas
sender=...	Bắt buộc khi ghi state

Ket qua \ape test\ (2 passed):

```
PS D:\Study\IS355\TH\w2> D:\Study\IS355\.venv-ape\Scripts\ape.exe test
===== test session starts =====
platform win32 -- Python 3.12.10, pytest-8.4.2, pluggy-1.6.0
rootdir: D:\Study\IS355\TH\w2
plugins: eth-ape-0.8.52, platformdirs-4.12.3
collected 2 items

tests\test_simple_storage.py .. [100%]

===== warnings summary =====
tests/test_simple_storage.py::test retrieve
  D:\Study\IS355\.venv-ape\Lib\site-packages\pydantic_settings\sources\utils.py:47: IncompleteFieldDefinitionWarning: Field 'import_remapping' has an incomplete definition: its annotation contains an unresolved forward reference, so setting s sources may fail to correctly resolve its value. Call 'model_rebuild()' on the model where the field is defined, once all the referenced types are defined.
    warnings.warn(

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 2 passed, 1 warning in 0.86s =====
```

Warning pydantic bo qua.

```
In [ ]: from pathlib import Path
out = Path('tests/test_simple_storage.py')
out.parent.mkdir(parents=True, exist_ok=True)
out.write_text('import pytest\n\n@pytest.fixture\ndef deployer(accounts):\n    re
print('Wrote', out.resolve())
print('Chay: D:\\Study\\IS355\\.venv-ape\\Scripts\\ape.exe test')
```